

iBEAM smart

Manual

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(December 2025 Subject to changes without notice)

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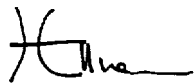
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As we are constantly improving our products, we greatly value all customer feedback. We encourage you to tell us what you like about our products as well as any suggestions for improvement.

Best regards,



Harald Ellmann
Director Service
TOPTICA Photonics SE

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1 General Description of the iBEAM smart Laser Source

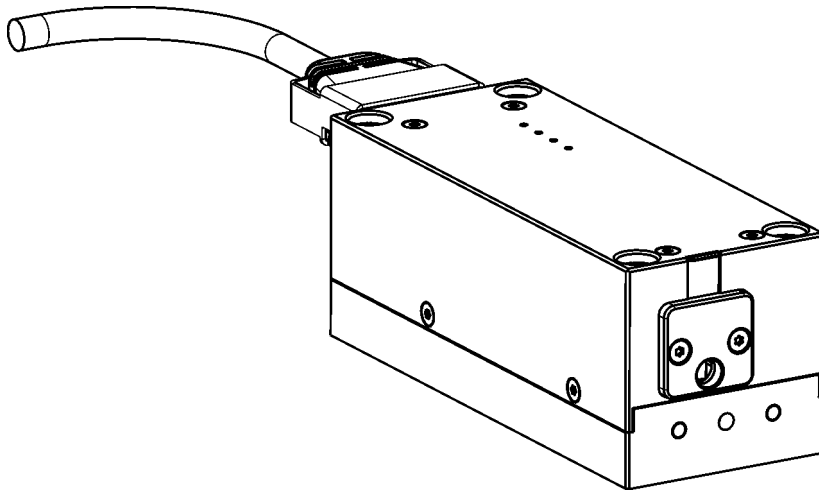


Figure 1 iBEAM smart OEM laser head

DANGER ! The iBEAM smart Laser Source is an OEM Product only.

CAUTION ! It is specially designed for the integration into the customer's main assembly. The operator or designer of the main assembly is responsible for integration of a **key switch and an inter-lock circuit**.

The iBeam smart represents the first true one-box solution in compact OEM diode laser modules. The small size, including all essential functions, is made possible by a highly sophisticated microprocessor based electronic setup in combination with an outstanding mechanic and optical design.

Key Features of the iBEAM smart:

- Highest available power of single emitter, single-mode diode lasers
- Diffraction limited, circular TEM₀₀ beam
- Best long-term power and pointing stability
- Active thermal control of laser diode for best stability and reliability (additional heat sink needed)
- Analog output control (analog modulation) up to 1 MHz
- Small-size, all-in-one OEM package
- High performance, low noise laser diode driver incorporating μ Processor and RS 232 control

Options:

- Pulse Option (digital output control)
- SMARTDOCK (Fiber Coupling)
- Optical Isolator (iBEAM smart WS)

1.1 iBEAM smart WS (Wavelength Stabilized)

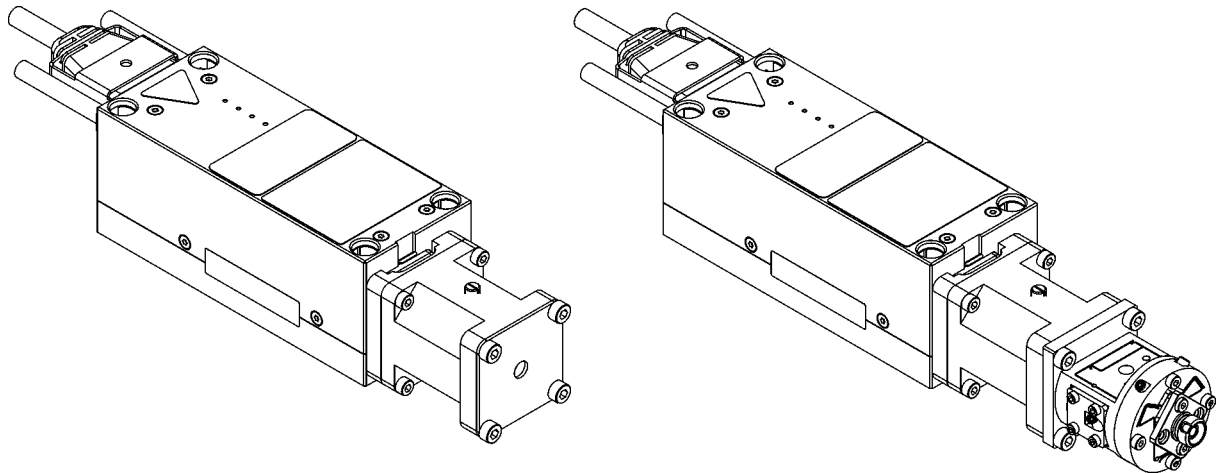


Figure 2 iBEAM smart WS with isolator option, free beam (left) and with SMARTDOCK option (right)

Instead of a standard Fabry-Perot laser diode, the iBEAM smart can also be equipped with a wavelength stabilized laser diode. Depending on the wavelength this laser diode can be a DFB, DBR or VHG stabilized semiconductor that emits single frequency laser light. The typical linewidth is in the range of 10 MHz-20 MHz.

In general, operation of the iBEAM smart WS is very similar to the standard iBEAM smart and all features are available and could be used. Please note that some of them (e.g., the pulse option or the FINE/SKILL features) might be in contrast to the overall goals when using a narrow linewidth laser system in an application.

There are two major differences compared to the standard iBeam smart:

- The specifications are only valid for the factory settings of the operation point (i.e., power, temperature etc.). Changing the power level or the temperature of the laser diode will result in a wavelength change and a potential unstable single frequency operation (multi-mode). This rule is especially important when a VHG laser diode is incorporated. DFB diodes are less critical and the power can actually be adjusted over a larger range while keeping single-frequency emission.
- Like all single-frequency lasers the iBEAM smart WS is very sensitive to optical feedback. In addition to the risk of destroying the laser diode itself, more importantly optical feedback will disturb the single-frequency emission and lead to an unstable optical spectrum and often also output power. The use of an optical isolator is strongly recommended. In case of fiber coupling the iBEAM smart WS using the SMARTDOCK (please refer to section 8.2) this is a necessity!

1.2 Setup of the iBEAM smart Laser Head

DANGER ! Do not look into the laser beam as the output can exceed the limits for class 1 specified by the US laws 21 CFR 1040.10 and 2 CFR 1040.11 and the Laser Safety Standards EN 60825-1:2014+A11:2021, IEC 60825-1:2014. Take precautions to eliminate exposure to a direct or reflected beam.

DANGER ! Do not open the iBEAM smart Laser Head. There are no user serviceable parts inside.

1.2.1 Polarization

The iBEAM smart can be provided in two different optical setups (standard and large beam diameter). The orientation of polarization is depending on this optical setup (variations are noted in the Test Data Sheet).

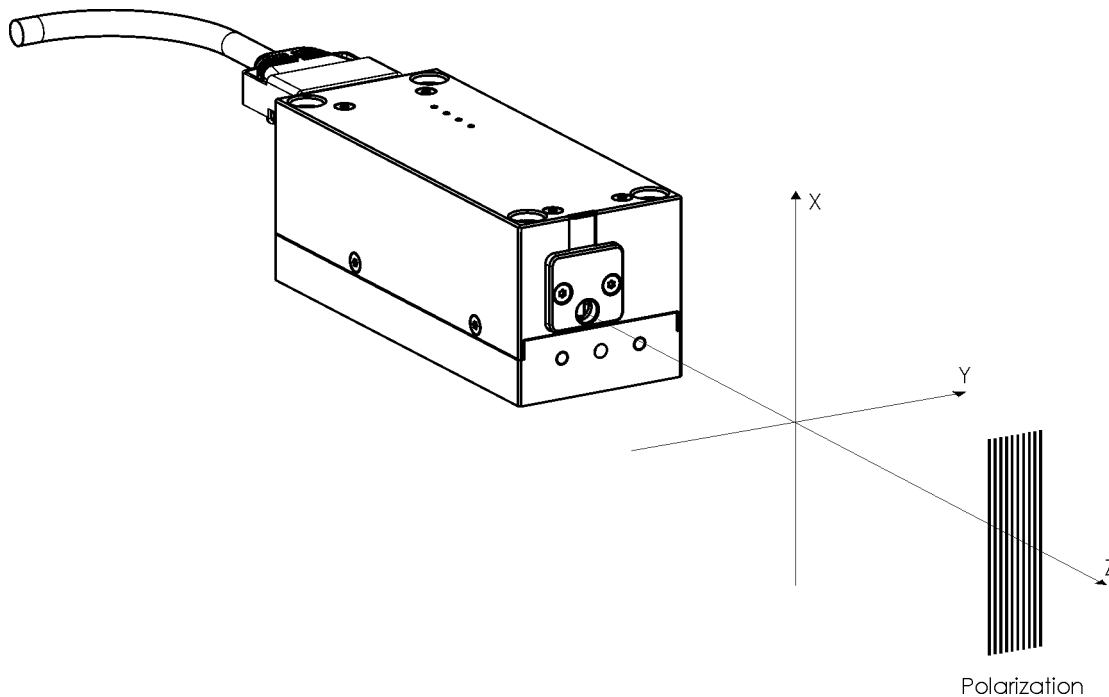


Figure 3 Standard beam diameter (approx. 1.3 mm $1/e^2$ dep. on LD and wavelength)

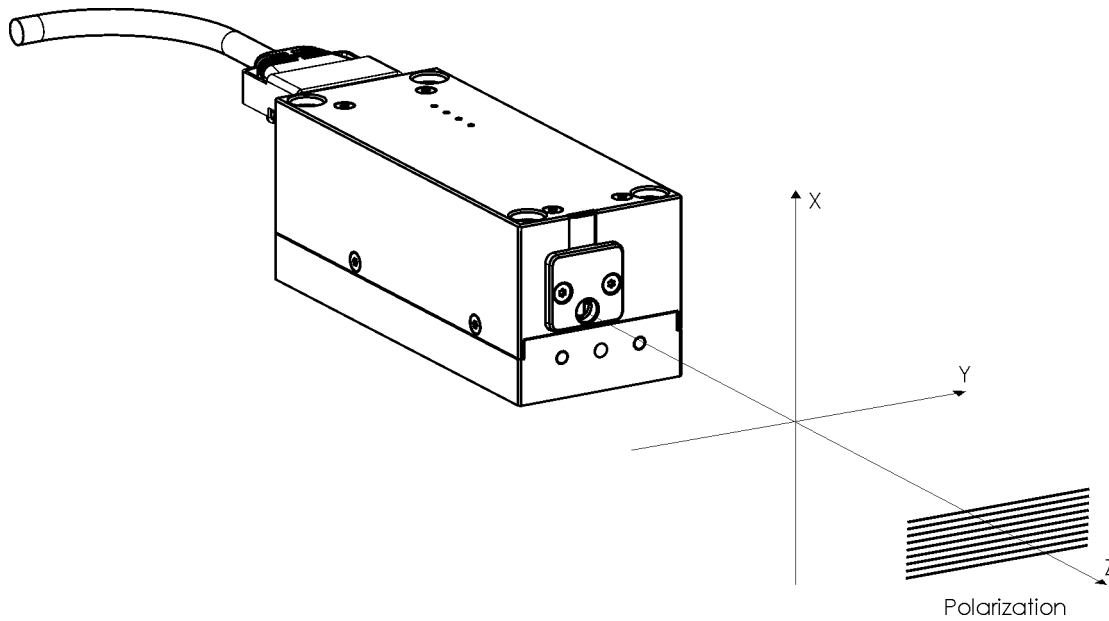


Figure 4 Optional large beam diameter (approx. 3 mm 1/e² dep. on LD and wavelength)

1.2.2 Thermal Control of the Laser Head

The whole collimator holder is thermally and electrically isolated from the environment, but nevertheless ruggedly fixed inside the housing. The iBEAM smart must be mounted onto an appropriate heatsink. Depending on the cooling temperature and the environmental temperature this heat sink needs to have good passive or active airflow around it. A thermistor is fixed next to the diode, which allows to measure the present temperature of the laser diode mount and thus active control of its temperature.

The whole setup guarantees that the unit laser diode and collimator, which needs to stay precisely aligned, has always the same fixed temperature, thus increasing the precision of the laser beam.

In the cool-down phase, when the iBEAM smart is switched on and the TEC driver works to stabilize the temperature as fast as possible, high currents are flowing through the TEC. The cool down procedure normally takes less than 5 minutes, in normal temperature conditions less than 1 minute.

1.2.3 Heatsink

The iBEAM smart must be mounted onto an appropriate heatsink. Maximum allowed heatsink temperature is 50° C(*) with 18 W heat dissipation from the laser module.

(*) With maximum laser diode power as of 2014/07/10.

1.3 Cable

The iBEAM smart is shipped with an RS 232 interface/interlock/power supply connection cable with attached interlock plug. For detailed description and installation please see section 3.

1.4 12 V DC Power Supply

The iBEAM smart must be supplied by a 12V DC ($\pm 5\%$) power source and uses a maximum peak current of 2 Amperes. A proper universal AC/DC Power Adaptor (100 - 240 V 47 - 63 Hz/12 V 5.0 A) can be ordered from TOPTICA Photonics SE

2 Safety Information

2.1 General Safety Terms

The iBEAM smart Laser Source is manufactured according to the International Laser Safety Standards EN 60825-1:2014+A11:2021, IEC 60825-1:2014 and complies with the US laws 21 CFR §1040.10 and §1040.11.

The following safety terms are used in this manual:

The **DANGER !** heading in this manual explains danger that could result in personal injury or death.

The **CAUTION !** heading in this manual explains hazards that could damage the instrument.

In addition, a **NOTE !** heading gives information to the user that may be beneficial in the use of the instrument.

DANGER ! Before operating the unit please read this manual carefully to prevent personal injury and
CAUTION ! damage to the device. The following safety instructions must be followed at all times.

DANGER ! The iBEAM smart Laser Source is an OEM Product only.

CAUTION ! It is specially designed for the integration into the customer's main assembly. The designer of the main assembly is responsible for the integration of a key switch into the 12 V DC Supply. An interlock circuit as well as all laser warning lamps and laser safety labeling must be implemented as well. In case the customer's main assembly is equipped with openings, where laser radiation can be accessed, the integration of a safety shutter may be required.

DANGER ! CAUTION !  **Possibility of electrical shock !** Wherever this symbol is attached, the possibility of an electrical shock may appear. Use only equipment and accessories supplied by TOPTICA.

 **Caution !** Wherever this symbol is attached read and understand the manual before operating the device. The manual must be consulted in order to find out the nature of the potential HAZARDS and any actions which have to be taken to avoid them.

DANGER ! CAUTION ! The iBEAM smart laser head is equipped with a LED that indicates laser emission. (Please refer to section 4.1 in this manual for detailed information). The system integrator of the iBEAM smart is responsible for mounting a second laser radiation emission warning light to guarantee redundancy. Be aware of laser emission when at least one of these LEDs lights up.

DANGER ! During installation, maintenance and service, all persons in the room must wear appropriate laser safety goggles while the laser is in operation. For a correct determination of the required laser protection goggles and other safety measures, please refer to section 2.2.

DANGER ! Laser safety goggles selected for adjustment purposes do not protect against an intentionally focused direct beam which will increase the optical power densities by a few orders of magnitude.

DANGER ! Regular functional checks and performance inspections at the supplier are recommended for all laser safety goggles.

DANGER ! The equipment must be positioned so that the disconnecting device can be easily reached by the operator.

DANGER ! Use of controls or adjustments or performance of procedures other than those specified herein may result in hazardous radiation exposure.

To be continued on the next page.

DANGER ! If the equipment is used in a manner not specified by the manufacturer, the protection provided by the equipment may be impaired.

DANGER ! Do not operate the iBEAM smart in a hazardous environment.

DANGER ! **Do not open the iBEAM smart Laser Head.** There are no user serviceable parts inside. Only authorized and specially-trained service personnel should perform internal tuning and replacement of components.

DANGER ! Do not look into the laser beam as the output can exceed the limits for class 1 specified by the US laws 21 CFR 1040.10 and 2 CFR 1040.11 and the Laser Safety Standards EN 60825-1:2014+A11:2021, IEC 60825-1:2014. Take precautions to eliminate exposure to a direct or reflected beam.

DANGER ! The iBEAM smart Laser Source can use very powerful laser diodes (up to class 3b). Therefore, it is imperative to take great care and observe the statutory warning labels on the unit.

DANGER ! Before connecting external supply units (e.g. computer, TTL pulses...) check all parameters carefully. Before connecting or disconnecting the iBEAM smart Laser Head to external units, disconnect the iBEAM smart Laser Head from the power supply.

CAUTION ! When using the iBEAM smart Laser Source, it is necessary to prevent any inadvertent electrical short-circuits of the Diode Laser and to avoid contact with current-carrying parts. The laser diode is particularly sensitive to voltage peaks. These might considerably affect the service life of the laser diode, even if a disturbance is not immediately noticeable. When making adjustments be sure to wear a high-impedance grounding strip around the wrist at all times.

CAUTION ! At all costs avoid high uncontrolled optical feedback into the laser diode, as this may damage or even destroy the laser diode. When making external adjustments during operation which may cause optical feedback, the controller current must be reduced to close above the lasing threshold. Only weakly reflecting objects may be allowed in the laser beam.

2.2 Determination of Laser Safety Goggles

DANGER ! During installation, maintenance and service, all persons in the room must wear appropriate laser safety goggles while the laser is in operation. For a correct determination of the required laser protection goggles and other safety measures, you should do the following:

- **1. Perform an Exposure Analysis**
- Consider where the exposure occurs (eyes, skin).
- Consider how long the exposure occurs:
 - Visible or invisible radiation?
 - Intended viewing or not? Viewing with unaided eye or with optical instruments?
- Consider the parameters of the irradiation at the location of exposure (*):
 - Power/power density and/or energy/energy density at the location of exposure.
 - Wavelength(s).
 - Beam diameter(s).
 - cw or pulsed radiation.
 - In case of pulsed radiation:
 - Determine the type of laser radiation: ultra-short pulses from a mode-coupled laser or other pulsed source?
 - Peak power and duration of the laser pulses
 - Time structure of the laser pulses (repetitive pulses, groups of pulses, single pulses)
 - Average power
 - Wavelength spread
- Consider any intentional actions, unintentional actions, foreseeable misuse and faults of the surrounding setup that could lead to a rise of the exposure.
- Consider technical and organizational safety measures at the workplace and training of the personnel.

2. Select Laser Goggles and other Safety Equipment

- The safety equipment has to safely reduce the maximum exposure determined in the previous step to levels below the legally allowed exposure limits at your workplace.
- Legally-binding exposure limits of employees at the workplace and the general public are in many countries specified in national laws and may differ from country to country.
Examples: OSHA regulations in the US, "Arbeitsschutzverordnung zu künstlicher optischer Strahlung - OStrV" in Germany.
- Maximum permissible exposures expressed as irradiance or radiant exposure and as power or energy can also be found in the (informative) Annex A of the laser safety standards EN 60825-1:2014+A11:2021, IEC 60825-1:2014.
Please note that the exposure limits in Annex A of EN 60825-1:2014+A11:2021, IEC 60825-1:2014 are not legal in nature and should be considered as a guideline only.
- Use only safety equipment that has been manufactured and approved for that purpose:
 - For example in the EU, use only laser safety goggles or laser adjustment goggles marked with a CE sign.
CE-marked goggles are tested against EN-208 and EN-207, the safety standards for laser eye-protectors and laser adjustment eye-protectors.
 - In other countries different regulations may apply !
- If more than one type of laser goggles satisfies the requirements, choose the one with the highest transmission for visible light (VLT).
- Do not overspecify the required optical density and protection level: laser goggles with too high protection levels may also reduce your sight and your viewing area and may thus lead to unwanted and unnecessary hazards.

(*) The irradiation at the location of exposure is not necessarily the same as the irradiation caused by the direct beam close to the beam aperture !

Depending on the optical setup and the location and diameter of the beam waist, the irradiation could be higher or lower compared to the irradiation at the beam aperture.

Even if access to the direct beam is impossible, consider things like stray light, reflected light and diffused light through materials and objects that might be intentionally or unintentionally located in the beam path.

After you have conducted an exposure analysis, please check which guidelines and laws are relevant for your location and institution for selecting the correct type of goggles. Below you can find an example as stated in the DGUV information 203-042 "Auswahl und Benutzung von Laser-Schutzbrillen, Laser-Justierbrillen und Laser-Schutzabschirmungen".

Beam Parameters of the Laser Product:

The beam parameters are similar across the different models of the iBEAM smart series. For the exemplary calculation of the necessary protective eyewear, the iBeam smart model with the highest power of the series is used.

Wavelength range: 370 .. 1060 nm
Output power: ≤ 0.5 W, CW
Beam diameter (1/e²): typ. 1 mm $\rightarrow d_{63}$ (1/e) = 0.71 mm
Divergence: ≤ 1.8 mrad

According to DGUV:

As the beam diameter d_{63} (1/e) is smaller than 1 mm, 1 mm has to be used as d_{63} diameter for the following calculation of the irradiance.

$$E = P/A_{63} = 0.5 \text{ W} / ((0.5 \times 1 \text{ mm})^2 \times \pi) \approx 6,37 \times 10^5 \text{ W/m}^2$$

Calculation of the correction factor:

$$F(d_{63}) = (d_{63}/\text{mm})^{1,7} = 1^{1,7} = 1$$

Leading to the corrected irradiance

$$E' = E \times F(d_{63}) = 6,37 \times 10^5 \text{ W/m}^2 \times 1 = 6,37 \times 10^5 \text{ W/m}^2$$

According to table 3 in DGUV 203-042 and based on DIN EN 207 this leads to a required protection level of LB5 for the laser safety goggles to be used with the system.

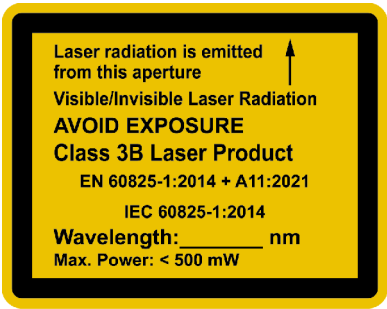
2.3 Safety Labels

2.3.1 Laser Beam Aperture

During operation, the laser beam is emitted through an aperture located on the front of the iBEAM smart Laser Head.

DANGER ! Through the aperture the laser system emits visible or invisible continuous laser radiation of up to 500 mW power (Class 3B laser product). Avoid exposing eyes and skin to the laser beam, including any laser stray light !

Depending on the specification of the iBEAM smart, the following labels or similar are affixed to the iBEAM smart housing according to EN 60825-1:2014+A11:2021, IEC 60825-1:2014:



Size: 38 mm x 30 mm
Color: Yellow/black
Location: Top of iBEAM smart Laser Head housing



Size: 10 mm x 10 mm
Color: Yellow/black
Location: 10 mm x 10 mm: iBEAM smart Laser Head housing right of fixed fiber
15 mm x 15 mm: Top of iBEAM smart Laser Head housing

2.3.2 Magnetically Sensitive Implants

Optical Isolators contain strong magnets. Magnetic fields extend beyond the protective housing of the device and may influence the operation of magnetically sensitive implants such as pacemakers or implanted defibrillators.

DANGER ! If you are wearing such implants, take care to keep away from the magnets. Always warn persons wearing such implants of the presence of the magnetic field. Persons with any magnetically sensitive implants should consult their medical doctor regarding any potential complications.

Depending on the specification, the following label is affixed on the iBEAM smart:



Size: 20 mm diameter
Color: Red/black
Location: iBEAM smart Laser Head housing

2.3.3 Magnets

The magnetic fields generated by optical isolators require special handling to prevent damage to the device or harm to persons. Do not attempt to disassemble the housing of the optical isolator.

DANGER ! The magnetic fields extend beyond the protective housing of the device and can attract
CAUTION ! other magnetic materials. Keep small ferromagnetic objects like tools, razor blades or screws away from the device. Keep your credit cards at a safe distance.

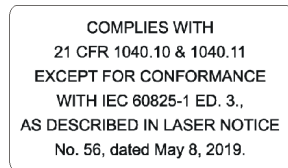
Depending on the specification, the following label is affixed on the iBEAM smart:



Size: 18 mm x 15 mm
Color: Yellow/black
Location: iBEAM smart Laser Head housing

2.3.4 CFR Compliance

Compliance with the US laws 21 CFR §1040.10 and §1040.11 is declared by the following label:



Size: 38 mm x 19 mm
Color: Silver/black
Location: iBEAM smart Laser Head housing

2.4 Identification of Manufacturer

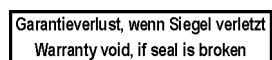
Manufacturer (name and address), production date, product ID number and compliance with CE standard are noted on the identification label. The crossed out wheeled dust bin symbol indicates that products must be collected and disposed of separately from household waste (please refer to section 10.11 for details).



Size: 38 mm x 19 mm
Color: Silver/black
Location: iBEAM smart Laser Head housing

2.5 Seal Label on Protective Housing

NOTE ! The protective housing of the iBEAM smart Laser Head is sealed by a label. Do not open the iBEAM smart Laser Head as this voids warranty.



Size: 35 mm x 8 mm
Color: White/black
Location: Outside of iBEAM smart Laser Head housing

2.6 Safety Features

2.6.1 Laser Beam Shutter

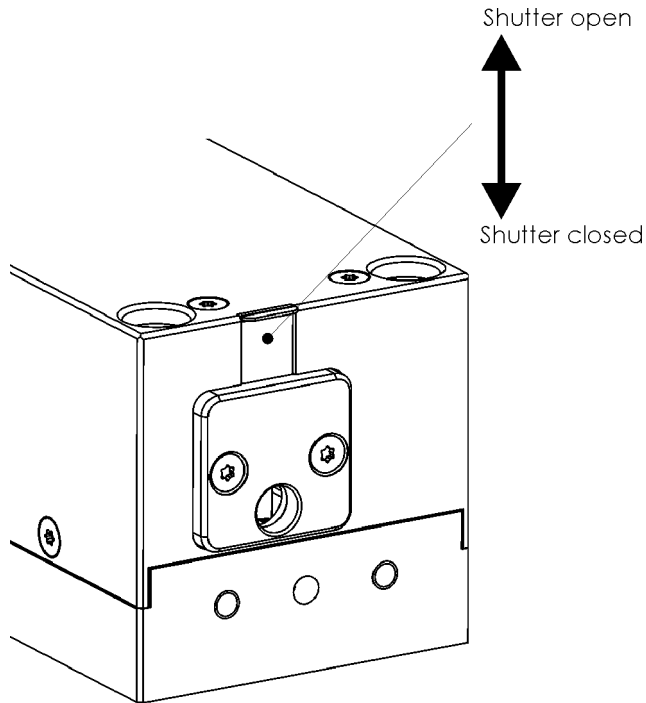


Figure 5 Laser beam shutter

Due to transport and laser safety the beam aperture of the iBEAM smart is protected by a shutter. To open/close the beam aperture, please shift the shutter as shown in Figure 5.

2.6.2 Interlock Plug

The iBEAM smart connection cable is equipped with an interlock connector (TYCO Type 0-0828, 2 pole) for installation of an interlock circuit. Alternatively the supplied interlock plug is used to close the interlock circuit.

3 Installation

3.1 Installation Instructions

DANGER ! The iBEAM smart Laser Source is an OEM Product only.

CAUTION ! It is specially designed for the integration into the customer's main assembly. The operator or designer of the main assembly is responsible for integration of a **key switch and an interlock circuit**.

The iBEAM smart set consists of the following parts:

- 1 iBEAM smart Laser Head
- 1 USB flash drive
- 1 RS 232 interface/interlock/power supply Connection Cable with Interlock Plug
- 1 iBEAM smart OEM-Manual (on demand)
- 4 Aluminium Mounting Screws M 4x12 mm

CAUTION ! Only use the supplied aluminium mounting screws ! Correct fixing torque is 1 Nm.

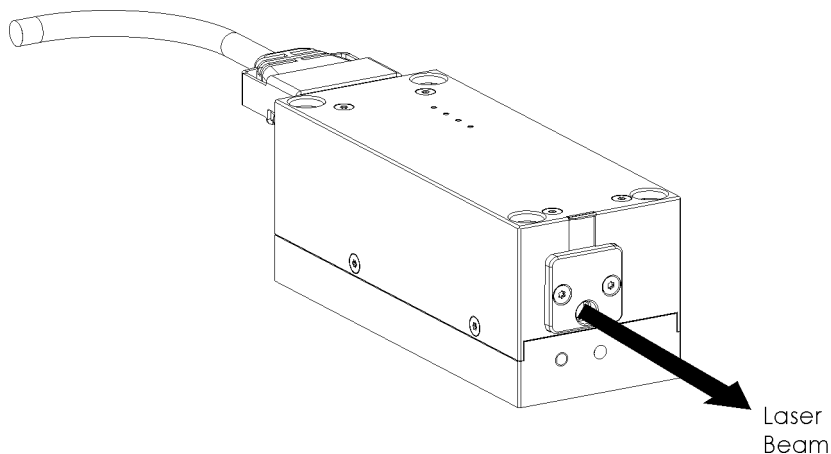


Figure 6 iBEAM smart laser head

When installing the iBEAM smart Laser Head the following instructions have to be observed:

- For general and environmental specifications, please refer to section 10.1.1.
 - The laser beam is emitted through the aperture (see Figure 6). For location of the beam exits, please see section 10.9.
 - The iBEAM smart Laser Head should only be installed at a place free from vibrations.
 - **Only use the supplied aluminium mounting screws ! Correct fixing torque is 1 Nm.** For main dimensions and location of the mounting screw bores please see section 10.9.
 - **The iBEAM smart Laser Head must be installed onto a suitable heatsink.** Maximum allowed heatsink temperature is 50° C(*) with 18 W heat dissipation from the laser module.
- (*) With maximum laser diode power as of 2014/07/10.

CAUTION ! The mounting surface of the heat sink must be flat to ensure good thermal contact. The mounting surface should be milled flat to better than 0.05 mm over the mounting surface.

NOTE ! The base plate of the iBEAM smart and the mounting surface of the heatsink have to be clean. Apply heat-conductive paste to the mounting surface of the heatsink.

CAUTION ! To guarantee maximum ESD protection the whole iBEAM smart setup must be grounded properly !

For main dimensions and mounting options of the iBEAM smart Laser Head please see section 10.9.

4 Operation

NOTE ! This section describes the general operation of the iBEAM smart laser system. While most of the description is valid for the iBEAM smart WS as well, please remember that some restrictions apply for this wavelength stabilized version (please refer to section 1.1)

4.1 Operator Controls

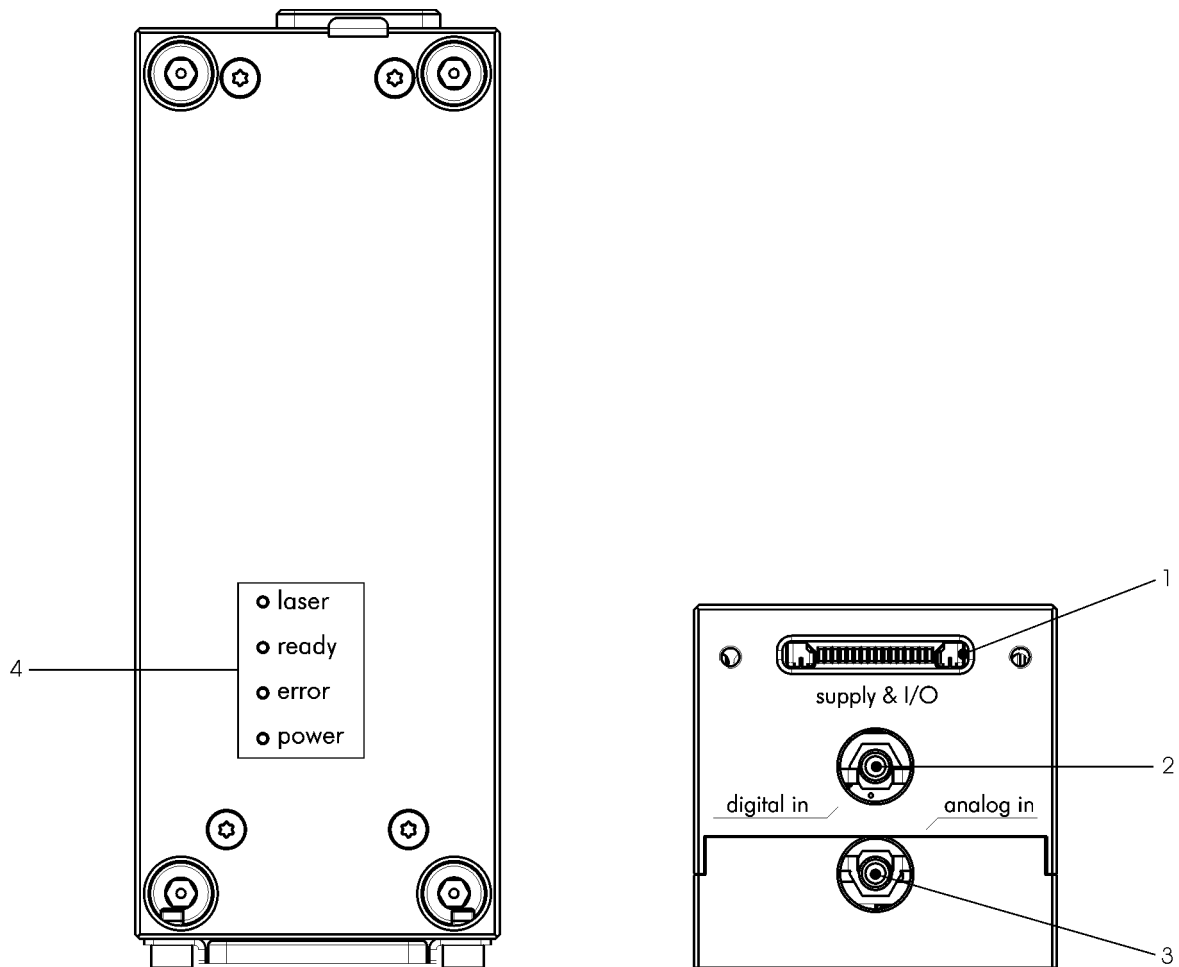


Figure 7 Operator controls of iBEAM smart laser head

- | | |
|---|-----------------------|
| 1 Supply & I/O connector | 3 Analog in connector |
| 2 Digital in connector (only with pulse option) | 4 Status LEDs |

1 Supply & I/O connector Molex Handy Link-connector	Connector for the power supply, interlock circuit and control of the iBEAM smart Laser Head by a computer. For pin assignment please see section 10.3.
2 Digital in SMB-connector	Only with pulse option. Frequency: up to 250 MHz Input Impedance: 50 Ω
3 Analog in SMB-connector	Analog Modulation Input. For pin assignment please see section 10.4. Frequency: up to 1 MHz Input Impedance: 5 k Ω
4 Status LEDs	The LED indicates: laser on: laser driver active blinking: overcurrent clipping has occurred. Reset clip by clicking Reset Clip in the TOPAS iBeam smart software or by sending the command reset clip. DANGER ! The laser LED serves as Laser Radiation Emission Warning Indicator. When the laser LED lights up, be aware that there is laser emission. ready on: TEC has stabilized. Ready for operation. blinking: TEC stabilizes. error error condition (for error codes please refer to section 9.5). power 12 V supply OK

4.2 Getting started with the iBEAM smart

CAUTION ! At all costs avoid high uncontrolled optical feedback into the laser diode, as this may damage or even destroy the laser diode. When making external adjustments during operation which may cause optical feedback, the controller current must be reduced to close above the lasing threshold. Only weakly reflecting objects may be allowed in the laser beam.

4.2.1 Laser Diode Basics

Laser diodes show an output characteristic as shown in the chart below. Increasing the supply current has very little effect on the power until the laser threshold is reached. As soon as lasing action has started, the “supply current to output power” shows a relatively linear dependence.

A laser diode is normally operated in this linear current to laser power range. This requires the bias current to be set to the laser threshold. Within this range the laser power can be modulated by the input current. This operation allows very high modulation speeds, but at the cost of modulation depth.

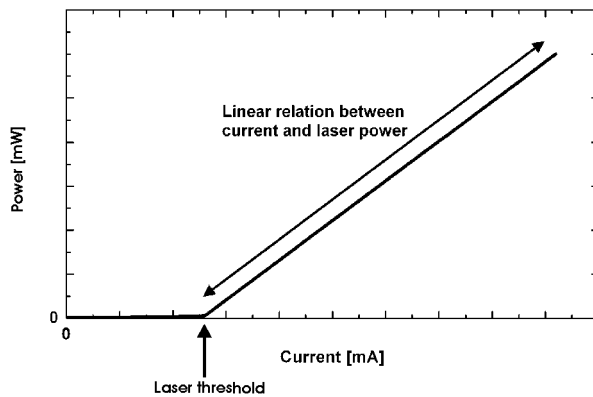


Figure 8 Linear laser current to laser power range and laser threshold

This means, that even in the “off” state (at or below the laser threshold), a very small amount of light is emitted. The so called “bias current”, which is constantly supplied to the laser diode, does generate some very low power light emission. In some applications this low power (bias) lasing is disadvantageous or even harmful. In these cases the bias current must be set to 0 so that the laser is operated in a “total off” mode, where no bias lasing occurs.

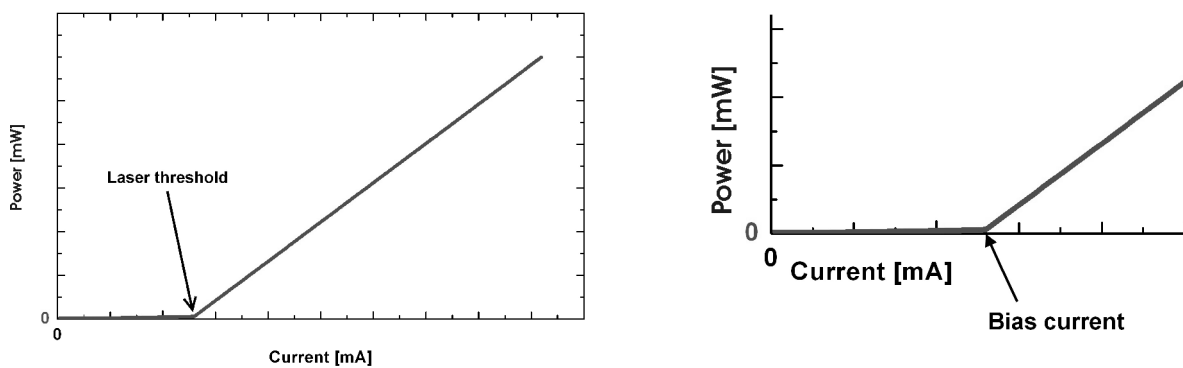


Figure 9 Laser threshold and bias current

4.3 iBEAM smart Basic Electronics Layout

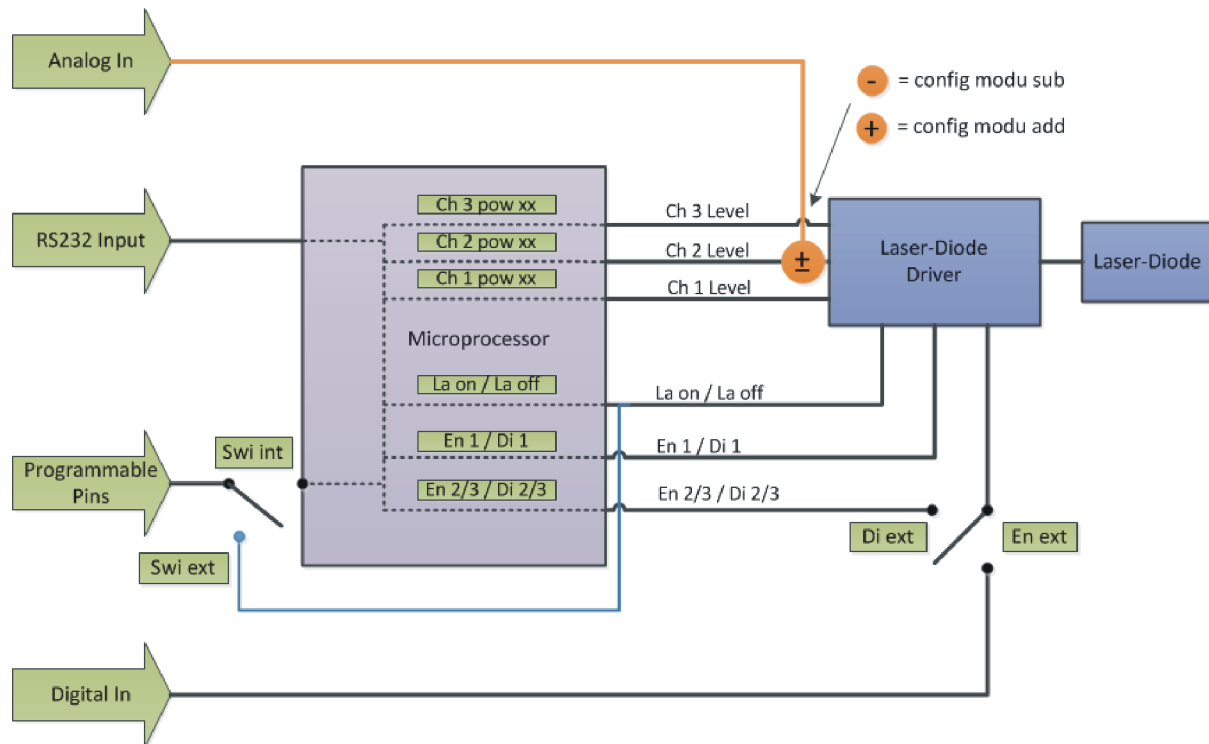


Figure 10 iBEAM smart basic electronics layout

The iBEAM smart control electronics consist of a special laser diode driver (LD-driver) and a microprocessor. The design is kept very flexible so that it can be configured to customer demands. There are a number of operation modes possible which are described in the following sections.

- Channel 1:** Bias level channel. Typically automatically activated with command la on. Behavior can be changed by software command (see section 9.4). Also needed for FINE operation (see section 6.1).
- Channel 2:** Normal operating power level channel.
- Channel 3:** Normal operating power level channel (only needed for special high power laser diodes or special applications).

4.4 iBEAM smart Operation Modes

4.4.1 Operation via RS232

This is the standard operation mode, the laser is fully controlled either by a terminal program (e.g., Windows™ HyperTerminal, PuTTY), a customer specific program using the RS232 commands for the iBEAM smart (please refer to section 4.8), or by using the TOPAS iBeam smart software (please refer to section 4.7).

If the control computer only features USB ports, the use of a RS232 to USB converter is necessary. For best compatibility, TOPTICA recommends to use converters with FTDI chips.

4.4.2 Operation via Programmable Pins (Internal Switches)

The iBEAM smart incorporates four programmable pins, that can be used to control the laser (please refer to section 7). RS232 commands can be allocated to these programmable pins. For example, three programmable pins could be used for:

- Laser ON and OFF (one pin)
- Enabling or disabling the external modulation (one pin)
- Setting two different power levels (one pin)

In this operation mode the microprocessor will “listen” to these programmable pins. When there is a change of the status of one pin, the microprocessor will execute the allocated command. The big advantage of operation via programmable pins is that they are completely configurable and can be inverted if needed. The drawback is that execution via the microprocessor limits the speed in which the commands can be executed.

4.4.3 Operation via Hardwired Pins (External Switches)

When hardwired pins are activated, the connection between the microprocessor and the laser diode driver is opened and the pins are directly connected to the laser diode driver (please refer to section 7.2). Operation via hardwired pins enables the commands to be executed much faster, because they are bypassing the microprocessor.

As the pins are now “hardwired” it is not possible to configure or invert them, the operation will be fixed to:

- Laser ON/OFF
- FINE ON/OFF

CAUTION ! In this configuration it is no longer possible to execute the functions Laser ON/OFF and FINE ON/OFF by RS232 software command, as the connection between microprocessor and laser diode driver is opened.

4.4.4 Operation via External Analog Modulation

Analog Modulation can be used to modulate the power set in channel 2 of the laser diode driver and can be configured in two ways, in “add” or in “sub” mode.

- “Add” means, that with increasing analog DC voltage the output power of the laser will increase (power is “added” with increasing analog DC voltage).
- “Sub” means, that with increasing analog DC voltage the output power of the laser will decrease (power is “subtracted” with increasing analog DC voltage).

For details please refer to section 6.3.

4.4.5 Operation via External Digital Modulation (Pulse Option)

In this operation mode the laser can be triggered via an external digital signal. As you can see in Figure 10, activating the digital modulation will switch the "en 2/di 2" pin from the microprocessor to the external TTL input (Digital In SMB-connector on the iBEAM smart rear panel). The laser can now be switched ON and OFF via an external TTL signal. The short rise and fall times of the iBEAM smart typically allow modulation bandwidths up to 250 MHz. The digital input signal can be completely asynchronous. The trigger level of the digital input line can be configured by software command (default setting: 1.3 V).

CAUTION ! Please be aware that when the external digital modulation is activated, it is not possible to enable or disable channel 2 by the RS232 command (en 2/di 2). Although the correspondent status change is set in the background, it only becomes effective after the external input line is de-activated by command dis ext.

4.5 Laser Beam Shutter

Due to transport and laser safety the beam aperture of the iBEAM smart is protected by a shutter. It must be opened before the first usage of the laser module. For operation of the shutter please see section 2.6.1.

4.6 Connections

After the mechanical installation is completed, please follow the steps below:

1. Connect the 9 pole female Sub-D9 connector of the RS 232 interface/interlock/power supply connection cable to a free serial port of your PC.
2. Use the interlock connector (TYCO Type 0-0828, 2 pole; for pin assignment please see section 10.5.3) of the RS 232 interface/interlock/power supply connection cable to set up an interlock circuit or use the supplied interlock plug.
3. Connect the Molex Handy Link connector of the RS 232 interface/interlock/power supply cable to the Supply & I/O connector of the iBEAM smart. The connector has a keying to allow only one correct orientation.
4. **It is recommended not to apply power to the iBEAM smart up to now.**

For standard operation via TOPAS iBeam smart software: Please proceed with section 4.7.

For standard operation via software commands: Please proceed with section 4.8.

4.7 Standard Operation via TOPAS iBeam smart

4.7.1 Installation of TOPAS iBeam smart

For TOPAS iBeam smart installation, please insert the supplied USB flash drive to the control computer and start the installer (TOPAS iBeam smart.exe). The installer will guide you through the installation process.

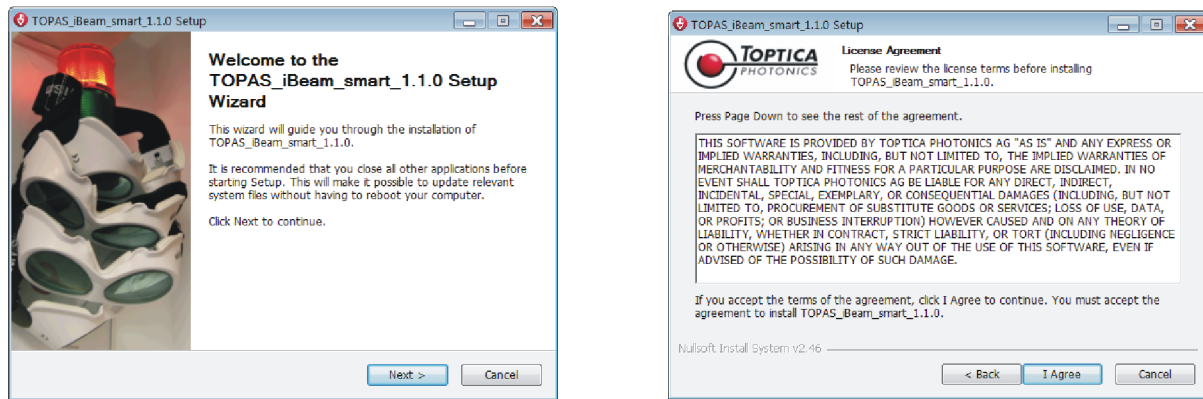


Figure 11 TOPAS iBeam smart installation (example)

Start the installation by clicking **Next** and confirm the license agreement by clicking **I Agree**.

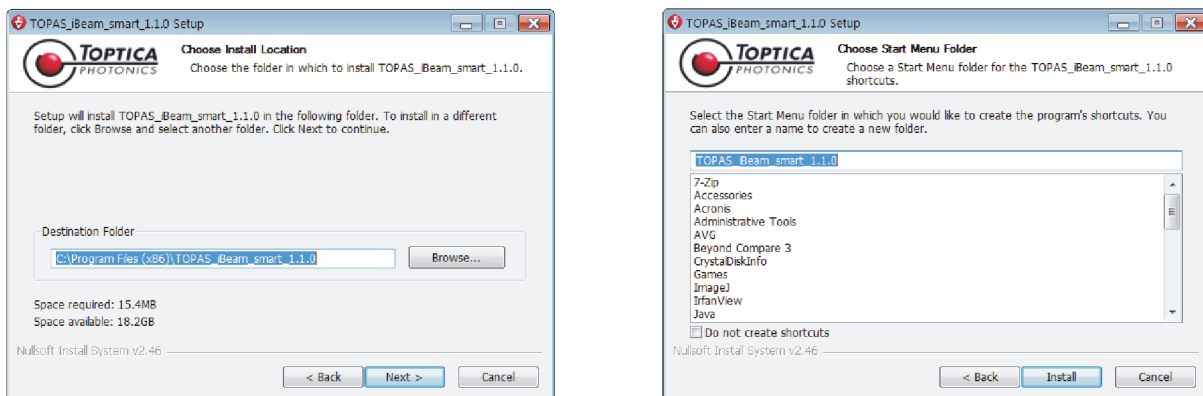


Figure 12 TOPAS iBeam smart installation (example)

Select the folder where TOPAS iBeam smart will be installed. Click **Next** to continue to the following window.

In your software version, all components of TOPAS iBeam smart will be installed. In the next window (Figure 12 right) you may choose the name of a program folder in the start menu. Click **Install** to continue.

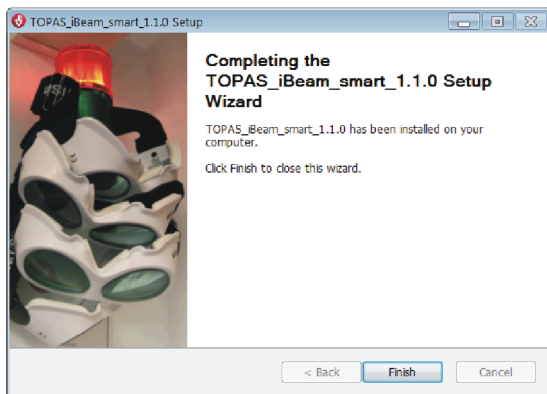


Figure 13 TOPAS iBeam smart installation (example)

After the installation is completed, click **Finish** to close the TOPAS iBeam smart installer.

4.7.2 Power Up and Boot Process

NOTE ! Please refer to section 4.1 for a detailed description of the operator controls on the iBEAM smart and to section 5 for a detailed description of the TOPAS iBeam smart operator controls.

1. Start TOPAS iBeam smart.
2. Connect the power supply to the 2.1/5.5 mm low voltage connector of the RS 232 interface/interlock/power supply connection cable. The iBEAM smart must be supplied by a 12V DC ($\pm 5\%$) power source and uses a maximum peak current of 2 Amperes.
3. Select **Menu > Connection Settings**

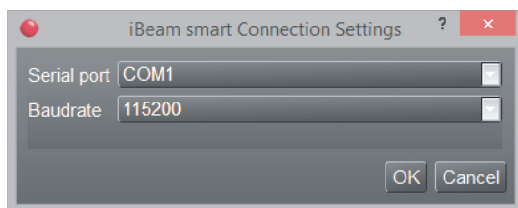
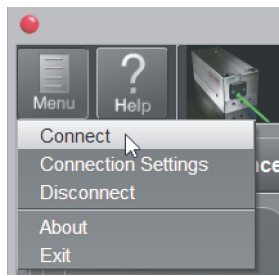


Figure 14 TOPAS iBeam smart Connection Settings window

4. When the Connection Settings window opens, specify the **Serial port** to which the device is connected. Click **OK** to save the settings and close the window.

**Figure 15**

5. Select **Menu > Connect** to establish a connection to the device.
6. The laser can now be operated with the Controls of the TOPAS iBeam smart software (see section 5).

The iBEAM smart will initialize all internal hardware components automatically for proper operation which means the levels are set to a pre-configured value. The laser diode is protected against over-currents and laser diode temperature will be controlled to a constant value.

DANGER ! For safe operation the laser is initialized but not switched on. Before switching on the laser please be very careful, don't look directly into the laser beam exiting at the aperture and observe the general safety instructions for lasers of class 3 b. The Safety Information in section 2 of this manual have to be followed at all times. Always wear appropriate laser safety goggles.

4.7.3 Enabling the Laser Output and Power Setting

The iBEAM smart is operated by the controls in the various tabs of the TOPAS iBeam smart software (see section 5). Enabling laser output and power setting is done in the Main or Advanced tabs.

4.7.4 Power Down

1. Switch off laser emission by clicking on the **Emission** button of the TOPAS iBeam smart software.
2. Select **Menu > Disconnect** to close the connection to the device.

Now you can disconnect the power supply adaptor.

4.8 Standard Operation via Software Commands

NOTE ! Please refer to section 4.1 for a detailed description of the operator controls on the iBEAM smart and to section 9 for a detailed description of the software commands.

4.8.1 Power Up and Boot Process

1. Please run your terminal program on your computer. The iBEAM smart will function properly using the 'Hyper Terminal' coming along with any release of the Microsoft® Windows XP® operating systems, but other terminal programs, such as PuTTY, should also work as long as the protocol parameters are set correct. Please see section 9.1 for configuring the terminal program (e.g. Windows HyperTerminal).
2. Connect the power supply to the 2.1/5.5 mm low voltage connector of the RS 232 interface/interlock/power supply connection cable. The iBEAM smart must be supplied by a 12V DC ($\pm 5\%$) power source and uses a maximum peak current of 2 Amperes.
The iBEAM smart should immediately power-up while sending a text information via the serial interface to the terminal program or iBP-Terminal which can be also found on the USB flash drive. This should look similar to the following example:

iBEAMsmart - Ultra Compact Diode Laser

(C)2009, TOPTICA Photonics AG

SerialN: iBEAM-SMART-488-S-A0-10188

PowerUP: 00008h

LaserON: 00006h

TECpass: USER

SysBoot: DONE

CMD>

If you do not see this screen please check cabling and/or terminal configuration and retry.

NOTE ! **When the laser does boot up but you do not see this screen**, please check the status LEDs. If all status LEDs are permanently on, please check for strong magnetic fields close to the beam exit.
A strong magnetic field (like an optical isolator) close to the laser beam output may inhibit the boot up process. In this case, please reduce the magnetic field, by increasing the distance between the magnetic field and the iBEAM smart. Then disconnect and reconnect the power supply to reboot the laser.

The iBEAM smart will initialize all internal hardware automatically for proper operation which means the levels are set to a pre-configured value, the laser diode is protected against over-currents and laser diode temperature will be controlled to a constant value. The proper initialization of this mode is confirmed by 'Sys Boot: DONE' on the terminal screen.

DANGER ! For safe operation the laser is initialized but not switched on. Before switching on the laser please be very careful, don't look directly into the beam and observe the general safety instructions for lasers of class 3 B. The Safety Informations in section 2 of this manual have to be followed at all times. Always wear appropriate laser safety goggles.

4.8.2 Enabling the Laser Output and Power Setting

The laser can be switched ON/OFF by the following commands:

CMD>	la on	activates LD driver
CMD>	la off	deactivates LD driver
CMD>	en [1..3]	enables specified channel Ch 1: bias level channel Ch 2: normal operating level channel Ch 3: only used at high-power models
CMD>	di [1..3]	disables specified channel

Example:

CMD> **la on**↵ (this is an abbreviation of 'laser on' which is also valid)

After boot up some light should exit at the aperture of iBEAM smart. To switch the iBEAM smart to the high (or exposure) level please enter the following command to activate channel 2 of the laser diode driver:

CMD> **en 2**↵ (this is an abbreviation of 'enable 2' which is also valid)

To switch back to the low (or bias) level please enter the command:

CMD> **di 2**↵ (this is an abbreviation of 'disable 2' which is also valid)

To change the output power of the high level please enter the following command.

CMD> **ch [x] power <level>**↵ for x enter the desired channel number
for <level> enter the desired power in mW

If you want to have these power levels to be recognized after resetting or power up of the system they must be stored into the EEPROM by the following command:

CMD> **save data**↵

4.8.3 Power Down

To switch off the laser please enter the command:

CMD> **la off**↵ (this is an abbreviation of 'laser off' which is valid also)

Now you can disconnect the power supply adaptor.

4.9 Getting System Information

4.9.1 General System Info

NOTE ! Only abbreviations instead of the whole commands are used in the examples !

There are several commands that will give you general information regarding the system:

CMD>	sh serial	shows the serial number of the module
CMD>	ver	shows the firmware version
CMD>	sta up	returns system and laser uptime

4.9.2 Actual System Info

NOTE ! Only abbreviations instead of the whole commands are used in the examples !

The following commands can be used to get info of the actual settings and status of the laser:

CMD>	sh cur	returns the actual LD current in cw operation
CMD>	sh limit	returns the actual setting of all limits. For a detailed description of the limits please refer to section 10.7.
CMD>	sh level power	shows the actual power setting for all available levels

4.9.3 Boot Configuration

Due to the very flexible μ -processor controlled electronics, several boot configurations can be set up and adapted to the customers need.

CMD>	boot on (default)	Activates standard boot mode. After system start the temperature stabilization is active, but the laser driver remains de-activated.
CMD>	boot all	in addition to the preset EEPROM values the following data are stored for initialization at system start: status of LD driver la on/la off status of channels channel settings

Besides this there are the several separate boot flags, that control only one special function

CMD>	boot fine	FINE feature is activated at system start
CMD>	boot pulse	0 - Autopulse feature not active after boot process. 1 - Autopulse feature activated right after boot process.

5 TOPAS iBeam smart Control Software

In this section all controls of the TOPAS iBeam smart graphical user interface (GUI) are described in detail.

5.1 Upper and Lower Screen Section

5.1.1 Header

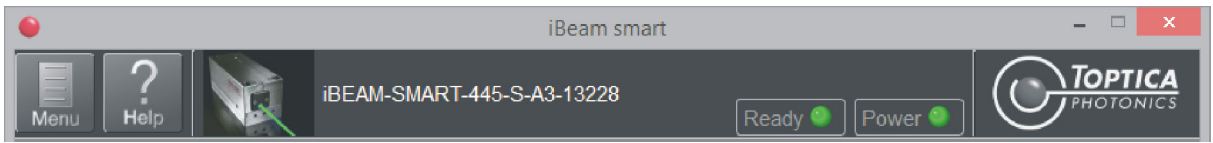


Figure 16 TOPAS iBeam smart header

Menu	Please refer to section 5.1.3.
Help	Please refer to section 5.1.4.
Connected Device	On top to the left the connected device and its serial number is displayed.
Indicators	Ready: green when TEC has stabilized. Ready for operation. red while TEC stabilizes. Power: green when power supply voltage is within specified range. red when power supply voltage is out of specified range (under-/overvoltage).

5.1.2 Footer

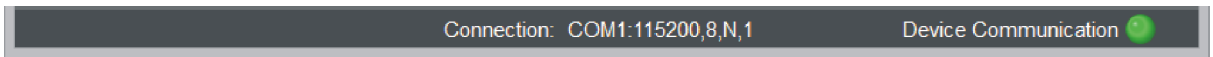


Figure 17 TOPAS iBeam smart footer

Connection	Displays details of the connection to an iBeam smart.
Device Communication:	Bright green: Communication. Dark green: No communication.

5.1.3 Menu

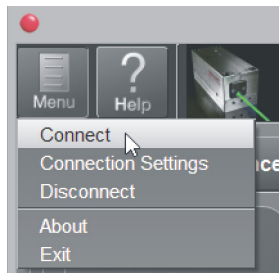


Figure 18 TOPAS iBeam smart Menu

Menu > Connect Establishes a connection to the device selected in the Connection Settings window.

Menu > Connection Settings The Connection Settings window opens where the connection can be configured.

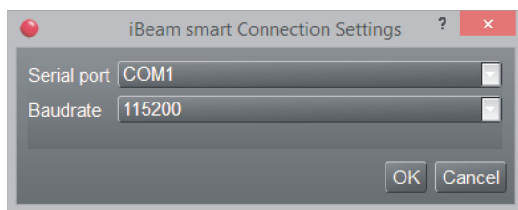


Figure 19 TOPAS iBeam smart Connection Settings window

Serial Port Select the **Serial Port** where the device is connected to.

BAUD rate For iBEAM smart lasers the BAUD rate is fixed to 115200.

OK Saves the settings and closes the window.

Cancel Closes the window without applying potential changes.

Menu > Disconnect Closes the connection to the device.

Menu > About The About window opens, which provides information about the manufacturer, contact details, software version, etc.

Menu > Exit Closes the TOPAS iBeam smart software.

5.1.4 Help

Clicking opens a pdf version of the iBEAM smart manual.

5.2 Control Section

5.2.1 Main Tab



Figure 20 TOPAS iBeam smart Main tab

Channel 1 Power

Slider

Set power level of channel 1.

Channel 1

Enable/disable channel 1 (bias level channel).

Input Field

Set power level of channel 1.
When the slider is moved, the display is changed accordingly.

Channel 2 Power

Slider

Set power level of channel 2.

Channel 2

Enable/disable channel 2 (operating power level channel).

Input Field

Set power level of channel 2.
When the slider is moved, the display is changed accordingly.

Temperature

Displays laser diode temperature.

Laser Section

Emission

Click the **Emission** button to switch laser emission on/off.

DANGER ! When laser emission is switched on, the green Laser Radiation Emission Warning Indicator in the **Emission** button lights up green. Be aware that there is laser emission.

Emission Warning Indicator

Green when laser driver is active.
Off when laser driver is inactive.

Digital Input

Only functional, if pulse option is installed.
Click **Digital Input** to activate the external digital input line.
Indicator Off: Digital input deactivated.
Indicator Green: Digital input activated.

Save Settings

Saves power settings, FINE settings and pulse settings to the EEPROM of the iBEAM smart.

Boot All

Saves a boot configuration to the EEPROM of the iBEAM smart. The current status of the device is restored at boot up.

Reset System

Resets the system (equivalent to switching off/on the power supply of the iBEAM smart).

Reset Clip

Resets the system in case of an error due to overcurrent clipping.

Status Section (display only)

Laser Clip

Green: OK.
Red: Overcurrent clip has occurred.

Temperature

Green: TEC set temperature reached.
Red: TEC set temperature not reached.

FINE

Off: FINE feature deactivated.
Green: FINE feature activated.
For details on the FINE feature please see section 6.1.

SKILL

Off: SKILL feature deactivated.
Green: SKILL feature activated.
For details on the SKILL feature please see section 6.2.

Pulse

Only functional, if pulse option is installed.
Green: Autopulse feature activated.
Off: Autopulse feature deactivated.

5.2.2 Advanced Tab

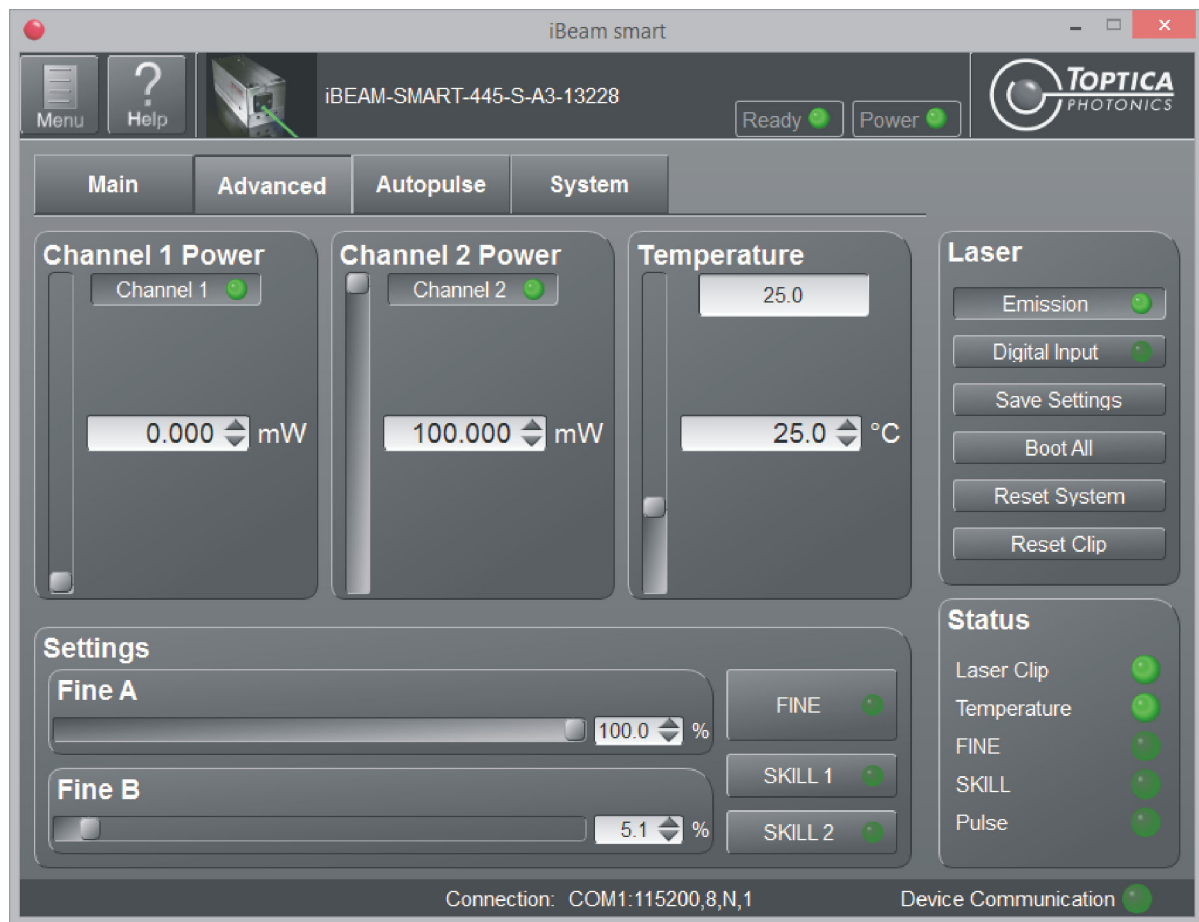


Figure 21 TOPAS iBeam smart Advanced tab

The Advanced tab allows to set the parameters for the FINE and SKILL features.
For description of the basic operator controls please refer to section 5.2.1.

NOTE ! You can only use either FINE **or** SKILL 1 **or** SKILL 2 at the same time.

Settings Section

FINE

Click to activate/deactivate the FINE feature.
For details on the FINE feature please see section 6.1

Fine A

Slider

Set FINE A parameter.

Input Field

Set FINE A parameter.
When the slider is moved, the display is changed accordingly.

Fine B

Slider

Set FINE B parameter.

Input Field

Set FINE B parameter.
When the slider is moved, the display is changed accordingly.

SKILL 1 Activate SKILL 1 settings.
For details on the SKILL feature please see section 6.2.

SKILL 2 Activate SKILL 2 settings.
For details on the SKILL feature please see section 6.2.

5.2.3 Autopulse Tab

NOTE ! The Autopulse tab is only available when the iBEAM smart is equipped with the Pulse Option.

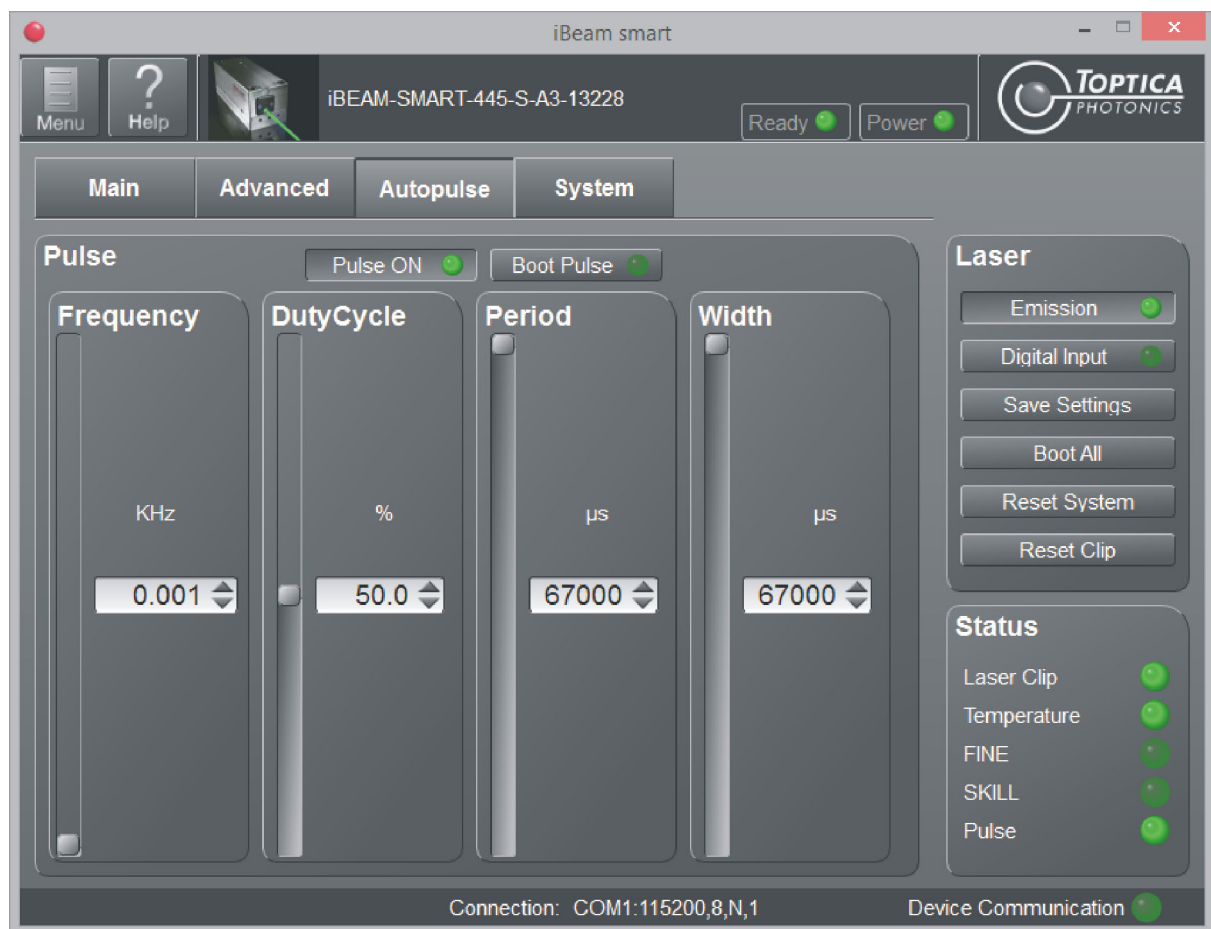


Figure 22 TOPAS iBeam smart Autopulse tab

The Autopulse tab allows to set the parameters for pulses generated by the iBEAM smart. For details on the Autopulse feature please refer to section 8.1.3.
For description of the basic operator controls please refer to section 5.2.1.

NOTE ! The Autopulse feature can only be used, if the external pulse input is deactivated (see section 8.1.1).
Autopulse only affects channel 2. Channel 2 must be disabled to use Autopulse as ch 2 enable overrides the Autopulse feature. If ch 2 is enabled, the laser will emit CW emission.

Pulse Section

Pulse ON	Enables/disables the Autopulse feature.
Boot Pulse	When clicked, the Autopulse feature is enabled automatically when the laser is switched on.
Frequency	Pulse frequency.
Slider	Adjust pulse frequency. When the slider is moved, the display is changed accordingly. Due to the adjustment also the settings of Period and Width change.
Input Field	Enter pulse frequency.
DutyCycle	DutyCycle is the laser on time in percents of a whole pulse period.
Slider	Adjust DutyCycle. When the slider is moved, the display is changed accordingly. Due to the adjustment also the settings of Width change.
Input Field	Enter DutyCycle.
Period	Length of a whole pulse period (laser on + off time).
Slider	Adjust Period. When the slider is moved, the display is changed accordingly. Due to the adjustment also the settings of Width and Frequency change.
Input Field	Enter Period.
Width	Width is the laser on time within a pulse period.
Slider	Adjust Width. When the slider is moved, the display is changed accordingly. Due to the adjustment also the settings of DutyCycle change.
Input Field	Enter Width.

5.2.4 System Tab

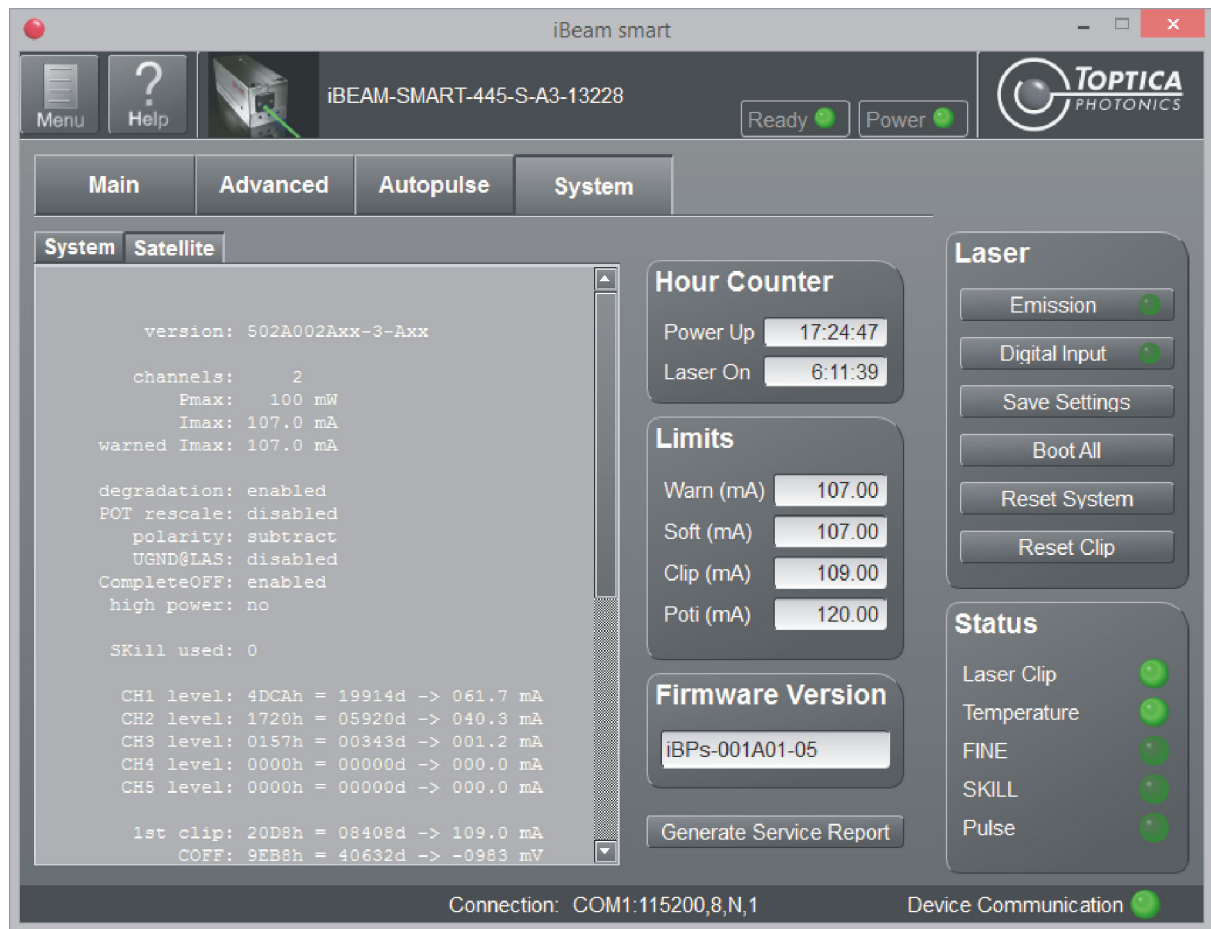


Figure 23 TOPAS iBeam smart System tab

The System tab provides detailed system information.

System Tab Displays the actual configuration stored in the iBEAM smart System EEPROM.

Satellite Tab Displays the actual configuration stored in the iBEAM smart Satellite EEPROM.

Hour Counter**Power Up**

Display the laser system on time (power on).

Laser On

Display the laser diode on time (emission on).

Limits**Warn (mA)**

Internal use only.

Soft (mA)

Internal use only.

Clip (mA)

Level of the (adjustable) LD protection circuit. If the current exceeds this level, the LD driver is switched off.

Poti (mA)

Level of the hardware configured clip (laser is switched off). The user can adjust this level. It defines the "end of life" of the laser diode.

Firmware Version

Displays currently installed firmware version.

Generate Service Report

Generates a file where actual configuration data is summarized.

6 iBEAM smart Features

The iBEAM smart comprises several additional features, which will be described in the following sections.

6.1 FINE

Power instabilities and high noise levels are common symptoms with any diode laser when there is a tendency for optical feedback in the set-up (e.g. back reflections from fiber end-facets or from highly reflective samples).

The Integrated Feedback Induced Noise Eraser can help to eliminate this problem. Varying the two FINE parameters will lead to a stable performance in most cases, making re-alignment and optical isolators unnecessary. For reasonable start values please see section 6.1.2.

Before the FINE feature can be effective, a major modification to the internal system configuration of the laser has to be done. The FINE functionality is tied to ch1 of the laser driver. As this channel is used as the "low"- or "bias"-level in the standard setup of the iBEAM smart, it is set to the lowest level possible. Therefore one would see nearly no effect when activating FINE. For proper operation of the FINE feature the laser diode must be at least partly driven via channel 1 of the LD-driver. The necessary changes are described in section 6.1.1.

NOTE ! You can only use either FINE **or** SKILL 1 **or** SKILL 2 at the same time.

6.1.1 FINE Configuration

In standard configuration, without FINE feature, the power of the laser beam is defined by two LD-driver channels, ch 1 and ch 2. Ch 1 represents the "bias level", whereas ch 2 defines the "operating power level" of the system.

In order to enable proper function of the FINE feature, at least part of the overall output power must be provided by using ch 1, as the FINE feature works off ch 1. The power assigned to ch 1 should equal about 60 mA drive current, which typically correlates to 30 - 50 mW for wavelengths < 500 nm, and ~ 5 mW for wavelengths > 500nm.

DANGER ! With the FINE feature activated, the laser system **emits laser radiation immediately after the command 'laser on'** because ch 1 is activated by default and does not have to be separately enabled. In contrast, ch 2 requires the command 'en 2'.

Example for Configuration via Software Commands:

CMD> **ch 1 pow xx**↵ (e.g. typ. 30 mW - 50 mW for wavelengths < 500 nm)

CMD> **ch 2 pow yyy**↵ (with yyy > xx)

Example for Configuration via TOPAS iBeam smart software:

Please note that FINE can also be configured in the Advanced tab of the TOPAS iBeam smart software (see section 5.2.2).

Enable **Channel 1** and set the **Channel 1 Power** to typ. 30 mW - 50 mW for wavelengths < 500nm, or ~ 5 mW for wavelengths > 500 nm).

Set the **Channel 2 Power** slider to a power level ≥ power level ch 1.

6.1.2 FINE Parameters

The following commands can be used to configure the FINE feature:

fine	off		feature deactivated
	on		feature activated
	a	<value>(0-100%)	setting of fine parameter "a". Recommended start value 90
	b	<value>(0-100%)	setting of fine parameter "b". Recommended start value 10

NOTE ! Before activating the FINE feature by cmd "fine on", it may be necessary to set the "fine a" to 0 to avoid clipping. Please try out.

Please note that the FINE parameters can also be set in the Advanced tab of the TOPAS iBeam smart software (see section 5.2.2).

6.2 SKILL

Speckle generation is an inherent effect of laser radiation. However, in many laser applications, especially imaging applications, where the a laser is used for illumination, speckle can be an unwanted effect.

iBEAM smart lasers include SKILL (Speckle Killer & Reduced Coherence), a feature which can reduce speckle pattern by reducing the coherence length of the emitted laser light. SKILL acts as a purely electronic "Speckle KILLer".

The SKILL function is a subset of FINE. While with FINE you can freely adjust the FINE parameters "A" and "B", SKILL works with two, fixed A & B parameter sets, namely SKILL 1 and SKILL 2. These presets have proven to be most effective to minimize speckle pattern in a majority of cases.

Before the SKILL feature can be effective, a major modification to the internal system configuration of the laser has to be done. The SKILL functionality is tied to ch 1 of the laser driver. As this channel is used as the "low"- or "bias"-level in the standard setup of the iBEAM smart, it is set to the lowest level possible. Therefore one would see nearly no effect when activating SKILL. For proper operation of the SKILL feature the laser diode must be at least partly driven via channel 1 of the LD-driver. The necessary changes are described in section 6.2.1.

6.2.1 SKILL Configuration

In standard configuration, without SKILL feature, the power of the laser beam is defined by two LD-driver channels, ch 1 and ch 2. Ch 1 represents the "bias level", whereas ch 2 defines the "operating power level" of the system.

In order to enable proper function of the SKILL feature, at least part of the overall output power must be provided by using ch 1, as the SKILL feature works off ch 1.

DANGER ! With the SKILL feature activated, the laser system **emits laser radiation immediately after the command 'laser on'** because ch 1 is activated by default and does not have to be separately enabled. In contrast, ch 2 requires the command 'en 2'.

Example for Configuration via Software Commands:

CMD> **ch 1 pow xx**↵ (e.g. typ. 15 mW - 25 mW for wavelengths < 500 nm)

CMD> **ch 2 pow yyy**↵ (with yyy > xx)

Example for Configuration via TOPAS iBeam smart software:

Please note that SKILL can also be configured in the Advanced tab of the TOPAS iBeam smart software (see section 5.2.2).

Enable **Channel 1** and set the **Channel 1 Power** to typ. 15 mW - 25 mW for wavelengths < 500nm, or ~ 2 mW for wavelengths > 500 nm).

Set the **Channel 2 Power** slider to a power level ≥ power level ch 1.

6.3 Analog Modulation

The iBEAM smart comes with an integrated analog modulation feature. Using the analog modulation, the optical output power can be varied with a bandwidth of up to 1 MHz by an analog DC voltage.

With the analog modulation, the analog DC voltage is fed directly to the Analog Input (3) at the iBEAM smart rear panel.

NOTE ! Analog modulation only affects channel 2 (ch 2) of the laser diode driver, so channel 1 (ch 1) should be disabled and set to 0 mW.

Input	Analog Input (3)
Modulation Range	≈ 0.01-100%
Max. Modulation frequency	up to 1 MHz (depending on modulation depth)
Input Voltage Range	0..5 V
Remarks	0 V nominal level 5 V maximum modulation depth

6.3.1 0 V = max. Power Configuration (Sub Mode, Default)

($V_{\text{analog}} \uparrow \rightarrow P_{\text{optical}} \downarrow$)

This is the default configuration of the analog modulation where the adjustment is performed starting with a high (mostly the maximum possible) optical power. It is selected by the CMD> **config modu sub** (see section 9.4).

By increasing the modulation voltage up to 5 V (other levels on request), the optical power is decreased down to a scalable minimum value. This minimum value is limited to some few microwatts due to necessary internal offset values. When the minimum value is set to above the laser threshold (typ. >1 mW, depending on the laser diode), a linear correlation between the analog DC voltage and the optical output power can be realized.

Scaling procedure:

1. Enter the system password (see section 9.2).

CMD> **pass system**

2. Set the laser to the desired maximum output power [mW]:

CMD> **ch 2 pow xx.**

3. Start the scaling routine:

CMD> **trim modu.**

4. Apply a high level analog DC voltage to the Analog Input (3) at the iBEAM smart rear panel. This voltage shall correspond to the maximum modulation depth, apply 5 V if you want to use the entire 0 - 5 V range.

NOTE ! Please do not exceed the maximum analog DC voltage (5 V as a standard)

(to be continued on next page)

5. Using the +/- keys on the computer keyboard, decrease/adjust the output power to the desired lower level which shall correspond to the maximum modulation depth (= minimum laser power output).
6. Leave the 'trim'-command by typing 's'
7. Store the direct analog modulation scaling to the EEPROM:

```
CMD> save modu↵  
CMD> save data↵
```

6.3.2 0 V = min. Power Configuration (Add Mode)

($V_{\text{analog}} \uparrow \rightarrow P_{\text{optical}} \uparrow$)

In this configuration the adjustment is performed starting with a low (mostly the lowest possible) optical power. It is activated by the CMD> **config modu add** (see section 9.4).

The minimum value is limited to some few microwatts due to necessary internal offset values. By increasing the modulation voltage up to 5 V (other levels on request), the optical power is increased up to a scalable maximum value (usually the maximum possible optical power). When the start value for the adjustment (minimum value) is set to above the laser threshold (typ. >1 mW, depending on the laser diode), a linear correlation between the analog DC voltage and the optical output power can be realized.

Scaling procedure:

1. Enter the system password (see section 9.2).

```
CMD> pass system
```

2. Set the laser to the desired minimum output power [mW]. This must be below the maximum output power, as in this configuration the power is increased by applying an analog DC voltage:

```
CMD> ch 2 pow xx↵ typically xx [mW] will be set to "0.001" (
```

NOTE ! For software reasons, xx must be set to 0.001 if the laser output power should be set to 0. If a more linear correlation between input voltage and laser power is desired, please set xx to laser threshold (approx. 1 mW).

3. Start the scaling routine:

```
CMD> trim modu↵
```

(to be continued on next page)

4. Apply a high level analog DC voltage to the Analog Input (3) at the iBEAM smart rear panel. This voltage shall correspond to the maximum modulation depth, apply 5 V if you want to use the entire 0 - 5 V range.

NOTE ! Please do not exceed the maximum analog DC voltage (5 V as standard)
--

5. Using the +/- keys on the computer keyboard, increase/adjust the output power to the desired upper power level which shall correspond to the minimum modulation depth.

NOTE ! The behavior is inverted, + will decrease the power, - will increase the power.

6. Leave the 'trim'-command by typing 's'
7. Store the direct analog modulation scaling to the EEPROM:

```
CMD> save modu↵  
CMD> save data↵
```

7 Programmable Pins

NOTE ! Password Level 1 required

The iBEAM smart is equipped with programmable pins which can be accessed via the Supply & I/O-connector at the laser, or, when using the supplied connection cable, via the RS232 connector. For pin assignments and voltage levels please see sections 10.3 and 10.5.2.

The four programmable pins are:

Name	Function	Label in the Software
Laser On	Laser on /off	SW1
FINE On	FINE on / off	SW2
Switch 1	Free	SW3
Switch 2	Free	SW4

7.1 Programmable Pins used as Internal (Software Defined) Switches

All pins can be used as internal switches. That means that the signals applied to the pins are wired to the microprocessor which will execute commands according to the applied signals. This allows free programming of the commands that are executed according to the pin input signals.

Example: How to configure Programmable Pins as Internal Switches:

`config swi int` configures programmable pins as internal switches.

NOTE ! To configure the programmable pins as external switches, please refer to section 7.2.

`config swi 1` 1 turns the internal switches on.
`Config swi 0` turns the internal switches off, i.e., all signals on the corresponding pins will be ignored, unless the switches are set to external (see section 7.2).

`save data` this will save the settings of the "config" command to the iBEAM smart, so that they will be active again after the next boot process. This command requires Password Level 3 !

`ini las` initializes the software so that the new settings can be applied.

How to configure the Internal Switches:

`swi x high` sets the selected switch to execute the chosen command at high level.
 x indicates the number of the switch SW1 – SW4)

`swi x low` sets the selected switch to execute the chosen command at low level.
 (x indicates the number of the switch SW1 – SW4)

`save swi` saves the configuration of the switches so that the configuration is available even after powering the laser off and on. This command requires Password Level 3.

7.2 Programmable Pins used as External (Hardwired) Switches

NOTE ! Only the functions Laser on/off and FINE on/off are possible.

The two pins "Laser on" and "FINE on" have a specialty: They can be configured as external switches, which means the signal applied to these pins will not pass the microprocessor, but will be directly wired to the laser diode driver. This direct wiring enables a much faster response (external switches: ~ 40 µs delay; internal switches: ~ 6 ms delay), but bears the drawback, that the RS232 commands la on/la off and fine on/fine off cannot be used anymore.

Thus, programmable pins configured as hardwired switches should only be used if the short trigger to light delay is needed.

NOTE ! External switches cannot be programmed, they are hardwired to the functions Laser on/off and FINE on/off.
External switches cannot be inverted.
When external switches are in use, their functions (Laser on/off and FINE on/off) cannot be controlled by the software commands via RS232 !

The following table shows how the programmable pins in external configuration are operated:

Name	Function	Low Level	High Level
Laser On	Laser on/off	Laser on	Laser off
FINE On	FINE on/off	Fine on	Fine off

Example: How to configure Programmable Pins as External Switches:

config swi ext configures programmable pins as external switches.

NOTE ! To configure the programmable pins as internal switches, please refer to section 7.1.

save data this will save the settings of the "config" command to the iBEAM smart, so that they will be active again after the next boot process. This command requires Password Level 3 !

ini las initializes the software so that the new settings can be applied.

NOTE ! After configuring the Programmable Pins as External Switches, the command "laser on" will no longer have an effect. Only using the "laser on" pin (PIN 1) at the RS 232 connector (see section 10.5.2) on the iBEAM smart will enable the laser driver. Same is true for the FINE feature (PIN 6) at the RS 232 connector. All other commands are not affected.

8 iBEAM smart Options

8.1 Pulse Option

The iBEAM smart pulse option enables the user to apply a digital pulse stream to the Digital In connector (2) of the laser, which then is directly converted into an optical output. An example is shown in Figure 24. In addition TOPTICA's "Autopulse" feature becomes available, which can make an external pulse generator redundant.

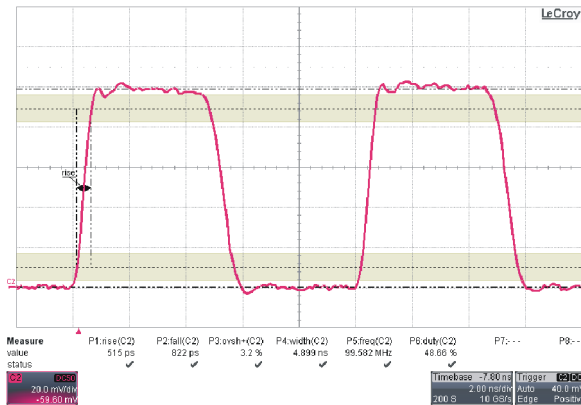


Figure 24 Digital pulse stream and optical output response

8.1.1 Standard Configuration of the Pulse Option

As a default the digital input path is not active. This enables the user to configure the laser and its power levels in cw-mode. When these settings have been finalized, the external digital input can be activated by

```
CMD> en ext.↵
```

Feeding a digital signal (the laser is configured for a 0 - 5 V TTL as standard, other configurations are possible) to the Digital In connector (2) will enable/disable channel 2 of the laser driver. A TTL high will lead to an optical output at a power level previously specified by

```
CMD> ch 2 pow xx.↵
```

A low signal at the digital input will disable channel 2. The optical output will be reduced to the remaining bias level of channel 1 (which can also be set to zero).

To disable the external digital input please enter

```
CMD> di ext.↵
```

After that the laser driver channels can be controlled as described in section 4.8.

8.1.2 Setting the Pulse Option as Default at Start Up

After the power levels have been configured once, it is possible to set up the system so that the external digital input is enabled right after a system boot. Thus it will be possible to start the pulsed operation immediately. The configuration can be stored as system default using the following procedure.

NOTE ! The iBEAM smart Laser System must be set at least to password level 1 (please refer to section 9.2 for Information).

CMD> **la on.**↵

CMD> **en ext.**↵

CMD> **boot all.**↵

8.1.3 Autopulse Feature

With the Autopulse feature the internal capabilities of the intelligent iBEAM smart electronics can be used. Frequencies up to 10 MHz can be programmed, the minimum pulse frequency is 1 Hz. The duty cycle can be set in the range from 1 to 99 % (depending on the frequency).

NOTE ! The Autopulse feature can only be used, if the external pulse input is deactivated (see section 8.1.1).
Autopulse only affects channel 2. Channel 2 must be disabled to use Autopulse as the command ch 2 enable overrides the Autopulse feature. If ch 2 is enabled, the laser will emit CW emission.

Commands:

pulse	off		Autopulse feature deactivated.
	on		Autopulse feature activated.
	freq	<value> (Hz)	Setting the frequency of the self-generated pulses. Inputs such as "1e6" for 1 Mhz are accepted.
	duty	<value> (1-99%)	Setting the duty cycle of the self-generated pulse stream.
boot	pulse	0	Autopulse feature not active after boot process
		1	Autopulse feature activated right after boot process

NOTE ! The system only accepts pairs of frequency and duty cycle settings that can be generated by the electronics.
When the settings cannot be processed, you will get the messages %SYS-E-040 parameter(s) and/or %SYS-W-041 out of range.

Please note that Autopulse can also be configured in the Autopulse tab of the TOPAS iBeam smart software (see section 5.2.3).

8.2 SMARTDOCK

The TOPTICA SMARTDOCK is a compact, rugged fiber coupler designed to be easy to use, while still having all the required degrees of freedom to allow maximum coupling efficiency to be achieved. The modular design allows a large range of lenses and standard fiber types to be accommodated within a single design. The monolithic construction results in improved long term stability when compared to traditional designs. By virtually eliminating mechanical coupling (cross talk) between the various alignment axes, as well as reducing hysteresis, a more intuitive and systematic approach to adjustment is facilitated, allowing higher coupling efficiencies to be achieved in shorter periods.

Please refer to section 8.2.2 for instructions on cleaning and handling the fiber.

8.2.1 Quickstart

NOTE ! This quickstart guide applies only to the iBEAM smart equipped with a SMARTDOCK and provided with a fiber. The use of a power meter is recommended for proper alignment.

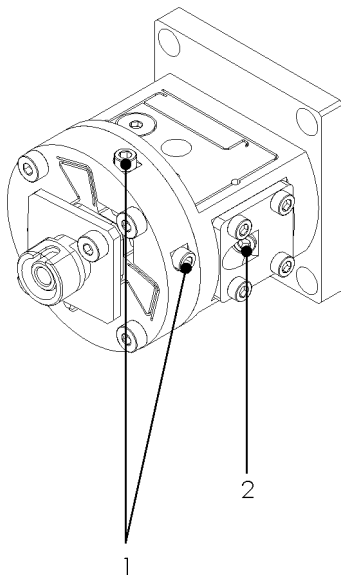


Figure 25 SMARTDOCK alignment

1. Connect the fiber to the SMARTDOCK and to the power meter. Switch ON the iBEAM smart.
2. Open the shutter as shown in Figure 5.
3. Using the X- and Y- fiber alignment screws (1), adjust the fiber position until the power meter reading is maximized.
4. Turn the lens focus adjustment screw (2) until the power meter reading is maximized.
5. Repeat steps 3 and 4 iteratively until the desired output power is reached.

8.2.2 Cleaning and Handling of the Fiber

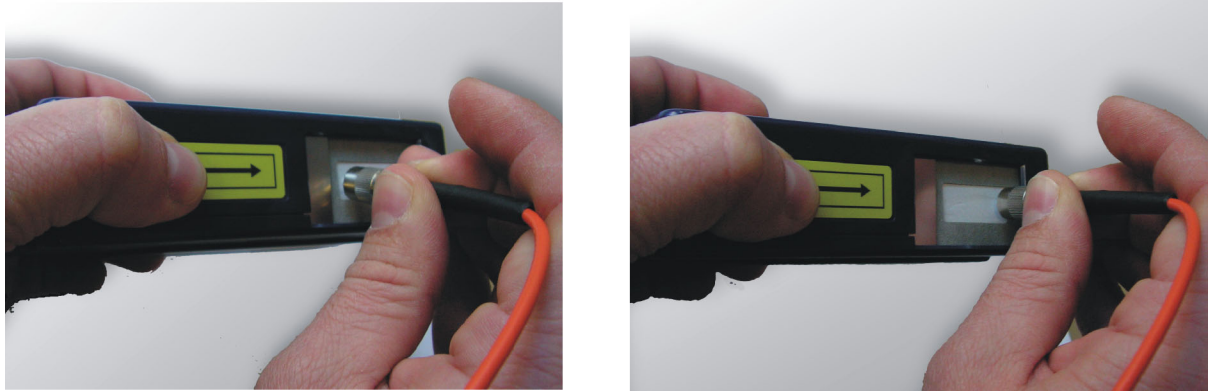


Figure 26 Cleaning the Fiber
Hold the Fiber as shown on the left side and move it twice across the cleaning surface as shown on the right side

Each time a fiber is connected, it must be cleaned with a commercially available 'automatic connector cleaner' or a similar device that incorporates a dry woven polyester cloth as cleaning medium (see Figure 26).

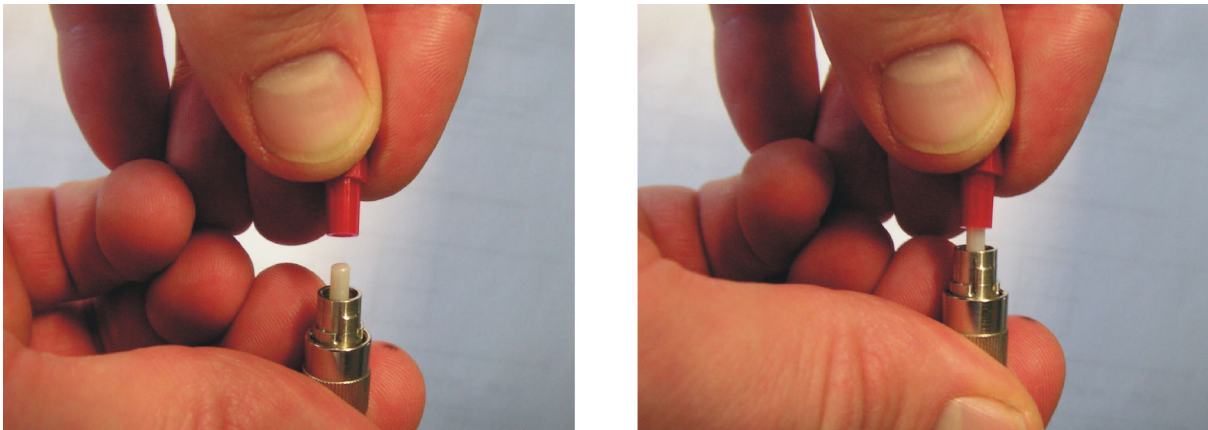


Figure 27 Protection of the Fiber during Installation

Carefully place the cap on the fiber end as shown in Figure 27 to protect it during installation or when not in use.

8.3 Isolator Option

For certain models of the iBEAM smart - mainly iBEAM smart WS with SMARTDOCK option - an isolator option is available. The optical isolator is integrated into a mount which can be attached to the front of the laser. Please refer to sections 10.9.3 and 10.9.4 for main dimensions of the iBEAM smart WS versions.

9 Software Control

9.1 Serial Interface Protocol

The parameters of the serial interface have to be set as follows:

Baudrate: 115200
 Data bits: 8
 Parity: none
 Stop bit: 1
 Handshake: none

The parameters of a terminal program (e.g. Windows HyperTerminal) have to be set as follows:

Serial Interface:

Direct connection via COMx with 115200,8,N,1 and serial interface handshake "none". That means that no hardware handshake (DTR, RTS) and no software handshake (XON,XOFF) of the underlying operating system is supported. Please use ANSI as terminal emulation and define backspace as Strg+H.

The prompt serves as a software-handshake. Before the system switches to the receive mode, it sends out a certain character string (the command prompt) with 'CMD>'. The input of further commands is then possible.

NOTE ! When the iBEAM smart Laser System is computer controlled, the command prompt always must be received after every command line before sending any further command lines in order to avoid the loss of characters or whole commands.

Command Structure:

Each command consists of printable ASCII characters (usually 'a-z' and '0-9'), which can be separated by one (or more) blanks (not at the beginning of a sequence of commands). Most of the pre-commands are divided into sub-commands. Whether a sub-command (or even a numeric parameter) must be entered depends on the individual command. For several commands the whole expression (as saved in the firmware) must be entered. Commands which are uncritical for the run may be shortened as long as they are clear (e.g. 'samp' can be shortened to 'sa', see section 9.3). All command sequences are case sensitive, i.e. no capital letters can be entered. Each command line ends with a 'carriage return' and a 'line feed'.

9.2 Password Level Structure

The iBEAM smart firmware incorporates a structured command set. Each command belongs to a specific password level. Thus commands that have deeper impact to the laser system setup are protected and cannot be entered inadvertently. Standard commands like 'la on/off' however can be entered without necessity of a password. The commands that are usually addressed by the customer are assigned to password level 0 and 1.

Level	Command	Password	Info
0	-	-	No password needed
1	pass service	TuiOptics	You can also enter level 1 by entering <i>!pass TuiOptics</i> directly
2	pass service 2	eibipi	
3	pass system	BlueNOTE!	

9.3 Serial Interface Commands Version iBPs-001A00-37 Level 0

Phrase 1	P1	Phrase 2	P2	Phrase 3	P3	Description
!pass TuiOptics						service password compatible to former PxLS
ch[annel]	[1..channel]int	pow[er]	[μW]float	mic[ro]		set power in μW for specified channel
ch[annel]	[1..channel]int	pow[er]	[mW]float			set power in mW for specified channel
di[sable]	0					disable all channels (including channel 1)
di[sable]	[1..5]int					disable specified channel
echo		off				disable echoing of characters sent from host within command entry
echo		on				enable echoing
en[able]	[1..5]int					enable specified channel
err[or]						show last error occurrence
help						show part of possible command specified by a mask
help		all				show complete list of possible commands for active password level
id						shows identifier or serial number (same as serial)
ini[tialize]		da[c]				initialize DAC and EPOT hardware from RAM table
int[tialize]		la[ser]				initialize laser
la[ser]		off				deactivate LD driver
la[ser]		on				activate LD driver
list		err[or]	[error]int (opt)			list error table or show error text if optional specified in command
fine on						enable FINE
fine off						disable FINE
fine a						setting FINE parameter a
fine b						setting FINE parameter b
pass [word]		reset				reset password level to locked state or normal command entry
pass [word]		serv[ice]	[2]int(opt)			service password (same as !pass TuiOptics) - value 2 enters 2nd service level
pass [word]		sys[tem]				system level password
pass [word]						show active password level
prom[pt]		off				disable prompt handshake to send only '[OK]' after each successful function call (otherwise an error message)
prom[pt]		on				activate prompt handshake (CMD>)
reset		clip				reset current clip
reset		sys[tem]				start system reboot
sa[mple]		cur[rent]				sample laser diode current continuously and show physical unit [mA]
sa[mple]		temp[erature]				sample laser diode temperature continuously and show physical unit [C]
save		data				save SAT and SYS areas from RAM into EEPROM hardware
save		sat[ellite]				save area from RAM into EEPROM hardware
save		sys[tem]				save area from RAM into EEPROM hardware
serial						returns S/N of the laser system

Phrase 1	P1	Phrase 2	P2	Phrase 3	P3	Description
sh[ow]		ch[annel]				show info of available channels
sh[ow]		cur[rent]				show actual LD current (cw)
sh[ow]		data				show RAM table contents
sh[ow]		flash				show flash parameters
sh[ow]		level		pow[er]		show power levels only from RAM table (extract of 'show sat')
sh[ow]		limit				returns actual setting of all limits
sh[ow]		pow[er]				returns actual laser power in μ W
sh[ow]		sat[ellite]				show RAM table contents
sh[ow]		serial				returns S/N of the laser system
sh[ow]		swi[tch]				show current assignments for program-able switches
sh[ow]		sys[tem]				show RAM table contents
sh[ow]		temp[erature]		sys[tem]		show base plate (heatsink) temperature
sh[ow]		temp[erature]				returns actual LD temperature
sh[ow]		tim[er]				returns system and laser uptime in seconds
sta[tus]		ch[annel]	[1..5]int			returns actual channel status (on/off)
sta[tus]		clip				returns actual status of the current clip
sta[tus]		la[ser]				returns actual status of the LD driver
sta[tus]		fine				returns actual status of the FINE feature
sta[tus]		soft				returns actual settings of the soft function
sta[tus]		supp[ly]				returns actual status of the supply voltages
sta[tus]		temp[erature]				returns actual status of LD temperature
sta[tus]		up				returns system and laser uptime in hours and seconds
talk		quiet				small textual output
talk		robot				absolutely no handshake (only requested messages by commands)
talk		usual				normal textual output
talk						show active talk level
trim[mer]		ch[annel]	[1..channel]int	rel[ative]		interactive trimming of laser diode current w/o initial change upon command entry
trim[mer]		pomic	[μ W]float	inc[rement]	[0..65535]int	interactive trimming of laser diode output power with power entry [μ W]
trim[mer]		pow[er]	[mW]int	inc[rement]	[0..65535]int	interactive trimming of laser diode output power with power entry [mW]
ver[sion]						returns version of firmware

9.4 Serial Interface Commands Version iBPs-001A00-37 Level 1

Phrase 1	P1	Phrase 2	P2	Phrase 3	P3	Description
baud	[9600..115200]int					set baud rate of I/O
bo [ot]		all				save the actual setup, which will be restored after a system reset
bo [ot]		check	[0..1]int			value 1 activates, value 0 disables PASS check
bo [ot]		flash	[0..1]int			value 1 activates and 0 disables the self generated flash option upon boot
bo [ot]		full				forces the laser to activate LD and enable ch2 after system boot
bo [ot]		off				TEC not active after system boot, DACs not initialized
bo [ot]		on				TEC active after system boot, DACs initialized
bo [ot]		tec	[0..1]int			value 1 activates, value 0 disables tec loop
ch [annel]	[1..channel]int	cur[rent]	[mA]float			set calibrated current level for specified channel and immediately apply it
conf [ig]		swi[tch]	[0..1]int			enable/disable switch functions from external digital signal (supply & I/O connector)
conf [ig]		swi[tch]		int		internal switch, wired through the micro-processor
conf [ig]		swi[tch]		ext		external switch, directly coupled to the laser diode driver (faster, but not configurable)
conf [ig]		ch	1	on/off		On: ch 1 will be automatically activated by the command "la on". Please note: ch 1 remains active even after laser is turned off by "la off". Off: ch 1 will not be automatically activated by the command "la on". If necessary (e.g., FINE operation), it must be activated by the user using command "en 1".
conf [ig]		modu		add[itive]		external analog modulation with additive behavior on laser diode current
conf [ig]		modu		sub[tractive]		external analog modulation with subtractive behavior on laser diode current
limit		clip	[mA]float			set current limit (hardware controlled)
limit		soft	[mA]float			set current limit (software controlled) used by some commands
list		pic				list P-I-Curve data
pic		shift	[mA]float			shift the PIC curve about specified current value

Phrase 1	P1	Phrase 2	P2	Phrase 3	P3	Description
sa [mple]		all				sample all signals connected to ADC and show raw data
sa [mple]		co[nt]				same as sample all but continuously for the most eight important signals
sa [mple]		co[nt]	sec[float]			same as sample continuous but with adjustable refresh rate from 0.1 to 6550 seconds
save		all				save all areas (memories) from RAM into EEPROM hardware (except backups)
save		flash				save area from RAM into EEPROM hardware
set		temp[erature]	[0..100]float	celsi[us]		set laser diode temperature in celsius and restart TEC controller (if previously closed)
swi [tch]	[1..4]int	high				set command to run, if external switch pin (at Supply & I/O connector) has a high level swi1: Las On swi2: FINE swi3: Switch 1 swi4: Switch 2
swi [tch]	[1..4]int	inv[ert]				invert polarity of switch specified (for connection list, enter show switch)
swi [tch]	[1..4]int	low				set command to run, if external switch pin (at Supply & I/O connector) has a low level swi1: Las On swi2: FINE swi3: Switch 1 swi4: Switch 2
swi [tch]	[1..4]int	norm[al]				normal polarity of switch specified (for connection list, enter show switch)
swi [tch]		reset				reset switch definitions to default
talk		gabby				enhanced textual output
tec		loop		close		close TEC loop
tec		loop		open		open TEC loop
trim [mer]		ch[annel]	[1..channel]int	cur[rent]	[mA]float	interactive trimming of laser diode current with current entry [mA]
trim [mer]		ch[annel]	[1..channel]int	da[c]	[0..65535]int	interactive trimming of laser diode current with DAC unit entry
trim [mer]		modu				interactive trimming of analog modulation strength at Analog Input

9.5 Error Codes

If an error occurs after an input, an error is displayed, e.g. '%SYS-W-003, missing'. 'W' represents the error type.

The error types are:

'I'	Information	e.g.: system ok (i.e. no errors detected)
'W'	Warning	e.g.: unknown (i.e. wrong command entered)
'E'	Error	e.g.: calibration (i.e. power/current calibration failed)
'F'	Fatal	e.g.: current limit (i.e. LD current exceeded max. limit)

With error type 'F' a digital signal (system error) is issued.

A complete list of all possible system messages can be found on the following page.

List of Error Codes

Error code 001 - 060

%SYS-I-000	no error
%SYS-I-001	no more errors
%SYS-E-002	error unknown
%SYS-F-003	system init
%SYS-F-004	system error
%SYS-I-005	system halt
%SYS-I-006	system resume
%SYS-I-007	system reset
%SYS-W-008	system active
%SYS-F-009	system calls
%SYS-F-010	low supply voltage
%SYS-I-011	aborted by system
%SYS-I-012	aborted by user
%SYS-E-013	not defined
%SYS-W-014	not supported
%SYS-F-015	null pointer
%SYS-E-016	conversion
%SYS-E-017	division by zero
%SYS-W-018	value negative
%SYS-W-019	overflow
%SYS-W-020	margin too small
%SYS-E-021	frame error
%SYS-E-022	overrun
%SYS-E-023	parity fail
%SYS-E-024	checksum error
%SYS-W-025	checksum false
%SYS-I-026	checksum done
%SYS-W-027	command unknown
%SYS-E-028	command not unique
%SYS-E-029	command missing
%SYS-E-030	command incomplete
%SYS-E-031	command token
%SYS-I-032	command cleared
%SYS-W-033	no digit
%SYS-W-034	too much digits
%SYS-W-035	digit required
%SYS-I-036	decimal
%SYS-I-037	hexadecimal
%SYS-W-038	too much chars
%SYS-E-039	input error
%SYS-E-040	parameter(s)
%SYS-W-041	out of range
%SYS-E-042	value missing
%SYS-E-043	password unknown
%SYS-E-044	password error
%SYS-E-045	password false
%SYS-I-046	password done
%SYS-W-047	access restricted
%SYS-I-048	access limited
%SYS-H-049	comport init
%SYS-E-050	comport scan
%SYS-F-051	device unknown
%SYS-F-052	device init
%SYS-E-053	device error
%SYS-E-054	device range
%SYS-E-055	DAC range
%SYS-E-056	memory unknown
%SYS-E-057	memory init
%SYS-F-058	memory error
%SYS-F-059	memory limit
%SYS-I-060	save done

Error code 061 - 105

%SYS-E-061	save failed
%SYS-I-062	not saved
%SYS-I-063	read done
%SYS-E-064	read failed
%SYS-F-065	sampler init
%SYS-E-066	sampler error
%SYS-W-067	sample clipping
%SYS-F-068	timer init
%SYS-E-069	system timer init
%SYS-E-070	signal unknown
%SYS-F-071	signal init
%SYS-E-072	signal error
%SYS-E-073	signal -> scale
%SYS-E-074	signal resolution
%SYS-E-075	scale error
%SYS-E-076	scale missing
%SYS-I-077	scaled
%SYS-W-078	unscaled
%SYS-E-079	calibration error
%SYS-E-080	calibration missing
%SYS-I-081	calibrated
%SYS-W-082	not calibrated
%SYS-E-083	inconsistent
%SYS-F-084	laser init
%SYS-E-085	laser error
%SYS-I-086	laser done
%SYS-W-087	laser tolerance
%SYS-I-088	laser loop openend
%SYS-I-089	laser on
%SYS-I-090	laser off
%SYS-W-091	laser degradation
%SYS-F-092	TEC init
%SYS-E-093	TEC error
%SYS-I-094	TEC done
%SYS-W-095	TEC tolerance
%SYS-I-096	TEC loop openend
%SYS-I-097	TEC on
%SYS-I-098	TEC off
%SYS-E-099	power too low
%SYS-W-100	power maximum
%SYS-W-101	power margin
%SYS-W-102	channel unused
%SYS-E-103	channel limit
%SYS-W-104	current limit
%SYS-E-105	current clip
%SYS-E-106	threshold missing

10 Appendix

10.1 Specifications

iBeam smart-S	375-S	395-S	405-S	420-S	445-S	460-S	473-S	488-S	505-S	515-S	633-S	640-S	660-S	685-S	785-S
Optical Specifications															
Wavelength* (nm)	375	395	405	420	445	460	473	488	505	515	633	640	660	685	785
Wavelength range (nm)	± 5	± 5	± 4	± 5	± 5	+0/-5	± 5	± 4	± 5	± 5	± 3	± 5	± 3	± 5	± 5
Output power (mW)	70	120	120/300	120	100	100	100	100/200	80	100	100	150/200	130	50	125/250
Min. fiber coupled output power (mW)	45	75	75/200	75	65	65	65	65/130	50	65	65	100/130	90	30	80/160
Power stability	< 0.5 % (drift over 48 h @ room temperature ± 5 °C)														
RMS noise	< 0.2 % (10 Hz - 10 MHz)														
Beam diameter (typ. @ 1/e²) (mm)	1.3	1.2	1.2	1.2	1.3	1.1	1.3	1.3	1.1	1.3	1	1.1	1.2	1.2	1,4/0.8
Beam shape	Circular (far field)														
Ellipticity	< 10 %														
Typ. divergence (mrad)	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6	< 0.6	< 0.7	< 0.7	< 0.8	< 0.7	< 1	< 1	< 1	< 1	< 1 / 1.5
M²	< 1.2														
Pointing stability	< 5 µrad/K														
Static alignment**	± 0.2 mm (xy), ± 0.5 mrad (angular)														
Polarization	Linear, Orientation: vertical (± 3°), Ratio > 100:1 (typ.)														
Electronic Specifications															
Digital Modulation (option)															
Supported signal levels	TTL														
Max. modul. frequency	250 MHz (100 MHz with complete off)														
Rise/fall time	< 1.5 ns (10 % - 90 %)														
Modulation extinction ratio	> 1000 : 1 (∞ with complete off)														
Analog Modulation															
Max. modul. frequency	1 MHz														
Analog modul. extinction ratio	10³														
Electronic Shutter															
Rise/fall time	40 / 10 µs														
Extinction ratio	∞ (complete off)														
General and Environmental Specifications															
Qualification	CE marked, Class IIb qualification, Level 4 ESD protection														
Digital interface	RS 232, ≤ 115,200 baud														
DC input	12 V DC, 2A														
Power consumption	< 18 W (typ. < 6 W)														
Heat dissipation	< 12 W (baseplate @ 50 °C)														
Warm-up time	< 5 min														
Temperature range	15 - 40 °C (operation), -10 - 60 °C (storage)														
Relative humidity	< 90 % (non-condensing)														
Dimensions	40 x 40 x 100 mm³ (H x W x D)														
*Other wavelength on request. **Static alignment tolerances are relative to reference holes in baseplate.															

*Other wavelength on request. **Static alignment tolerances are relative to reference holes in baseplate.

All specifications are subject to change without notice.

NOTE ! Specifications are subject to changes without notice ! For latest updates, please refer to the specifications on the TOPTICA website.

10.1.1 General and Environmental Conditions

- DC input requirements: 12 V (please refer to section 1.4).
- Power consumption: 18 W max. (6 W typ.).
- Environmental requirements: Indoors, altitude < 2000 m.
- Temperature Range:
 - Operating: +15 °C.. +40 °C, rel. Humidity non condensing with laser diode TEC at set point.
 - Storage: 0 °C.. +40 °C, rel. Humidity non condensing.
 - Transport: -10°C.. +60°C, rel. Humidity non condensing.
 - For a dew point table please refer to section 10.8.
- Weight: < 250 g.
- Size: approx. 40 x 40 x 100 mm³

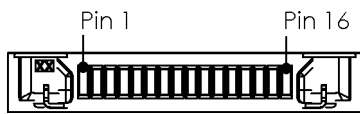
10.2 Cleaning

DANGER ! Before cleaning the iBEAM smart, switch off the iBEAM smart and disconnect the power supply from the mains line.

Clean the outside of the iBEAM smart with a cloth slightly moistened with a mild detergent solvent. Then dry with a clean cloth.

CAUTION ! If any materials or liquids get into the iBEAM smart during cleaning, contact TOPTICA. Never use strong solvents such as thinners, benzine, acidic- or alkaline solvents, spray-type cleaners or abrasive cleaners as they may destroy the surface.

10.3 Pin Assignment Supply and I/O Connector



1 Supply & I/O Connector						
Pin	Function	Dir	Z [Ω]	Voltage [V]	Current [mA]	Description
1	Signal GND			0		
2	Interlock					Interlock closed: short to GND
3	Supply + 12 V			12 ± 5 %	max. 2000	Supply Voltage
4	Supply + 12 V			12 ± 5 %		Supply Voltage
5	Return			0		Supply Voltage GND
6	Return			0		Supply Voltage GND
7	RS 232 TXA					RS 232 Interface Transmit
8	RS 232 RXA					RS 232 Interface Receive
9	Laser ON	Input	11k	0-5		Laser On, open collector input (= swi1 in the programming software)
10	FINE ON	Input	11k	0-5		Enable FINE feature, open collector input (= swi2 in the programming software)
11	Switch 1		11k	0-5		AUX Input (configurable) (= swi3 in programming software)
12	Switch 2		11k	0-5		AUX Input (configurable) (= swi4 in programming software)
13	Error	Output	200		10	For further information, please check with TOPTICA.
14	Ready	Output	200		10	For further information, please check with TOPTICA.
15	Monitor	Output	200		10	For further information, please check with TOPTICA.
16	Signal GND					

10.4 Pin Assignment SMB-Connector

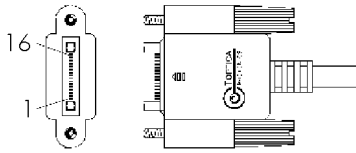
3 Analog in SMB-Connector						
Pin	Function	Dir	Z [Ω]	Voltage [V]		Description
1	Analog	Input	$\geq 5k$	0...+5		Direct analog modulation of LD current.
Case	Ground	-	-	-		Ground

3 Digital in SMB-Connector (only with pulse option)						
Pin	Function	Dir	Z [Ω]	Voltage [V]		Description
1	Digital	Input	$\geq 50 \Omega$ *	0...+5		Digital control of ch 2 status (ON/OFF).
Case	Ground	-	-	-		Ground

* Can be configured to higher impedance (e.g., 10 kΩ) at the factory.

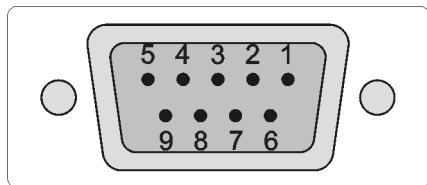
10.5 Pin Assignment of Connectors at Supply Cable

10.5.1 Supply & I/O Connector



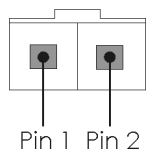
1 Supply & I/O Connector			
Pin	Function	Color at Cable	Description
1	Signal GND	black	
2	Interlock	brown	Interlock closed: short to GND
3	Supply + 12 V	red	Supply Voltage
4	Supply + 12 V	orange	Supply Voltage
5	Return	yellow	Supply Voltage GND
6	Return	green	Supply Voltage GND
7	RS 232 TXA	blue	RS 232 Interface Transmit
8	RS 232 RXA	purple	RS 232 Interface Receive
9	Laser ON	gray	Laser On, open collector input (= swi1 in the programming software)
10	FINE ON	white	Enable FINE feature, open collector input (= swi2 in the programming software)
11	Switch 1	pink	AUX Input (configurable) (= swi3 in programming software)
12	Switch 2	black/ white	AUX Input (configurable) (= swi4 in programming software)
13	Error	light blue	not available with standard cable
14	Ready	brown/ white	not available with standard cable
15	Monitor	orange/ white	not available with standard cable
16	Signal GND	red/ white	

10.5.2 RS 232 Sub-D9 Connector



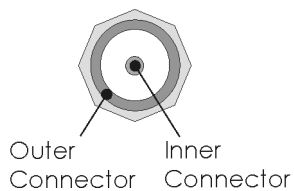
Pin	Function	Color at Cable	In programming software
1	Laser ON	gray	swi 1
2	RS 232 TXA	blue	
3	RS 232 RXA	purple	
5	Signal GND	red/white	
6	FINE ON	white	swi 2
8	Switch 1	pink	swi 3
9	Switch 2	black/white	swi 4

10.5.3 Interlock Connector TYCO Type 0-0828



Pin	Function	Color at Cable
1	Signal GND	black
2	Interlock	red

10.5.4 Power Supply Connector 2.1/5.5 mm Low Voltage



Pin	Function	Color at Cable
inner	Supply + 12 V	black
outer	Return	red

10.6 How to configure iBEAM smart for use with the Switchbox

Please attach the iBEAM smart to a computer which is equipped with a terminal program. You can also use the TopControl software to perform these steps, but please note that this program requires Windows XP or the Windows XP compatibility mode on Windows 7 systems.

When you have established a terminal connection, the following commands are required for configuration:

CMD> pass system Enter the service mode, needed for changing the switch settings.

PSW> BlueNOTE! Enter the password BlueNOTE!

%SYS-I-046, password done Answer from laser indicating that the password has been accepted.

CMD> swi 1 low Configuring switch one for low state.

STR> la on Command to be executed at low state.

CMD> swi 1 high Configuring switch one for high state.

STR> la off Command to be executed at high state.

CMD> swi 2 low Configuring switch two for low state.

STR> fine on Command to be executed at low state.

CMD> swi 2 high Configuring switch two for high state.

STR> fine off Command to be executed at high state.

CMD> config swi int Configure switches as "internal" switches. This means, that the state of the input pins (low or high) is transferred to the microcontroller, which will execute the entered command.

%SYS-I-109, re-init laser Answer from laser, indicating that the controller needs to be reinitiated.

CMD> ini la Initiate laser.

CMD> config swi 1 This command turns on the switch operation.

CMD> save swi This command saves the switch commands.

CMD> save data This command saves the configuration.

CMD> en 2 This command enables channel 2 (in case you are using channel 1 for your application, please substitute with "en 1", in case you are using channel one and two, please enable both, "en 1" and "en 2").

CMD> boot all Saves the enabled channels for the next boot operation. This configuration means that the laser power will start once you type "la on" or when you flip the laser on switch of the switchbox – no more separate "en 2" or "en 1" command is needed.

With these commands, your iBEAM smart is configured for the use with the switchbox.

10.7 Changing Current Limits

NOTE ! Only abbreviations instead of the whole commands are used in the examples !

There are 3 limits which can be set within the iBEAM smart software. As mentioned above the actual setting can be displayed by:

CMD>	sh limit	returns the actual setting of all limits. SOFT: x.x mA CLIP: x.x mA POTI: x.x mA
------	----------	---

- The "soft limit" (limit soft) only is an internal software limit. It prevents the system to run into the "current clip" especially when changing the LD current using the "trim"-command.
- The "current clip" (limit clip) is a real current clip and the laser is shut down if this limit is reached during operation. Due to potential aging of the LD after some hundred/thousand hours changing the limit might be necessary.
- The "poti clip" (limit poti) is factory preset clip and cannot be changed by the customer. It defines the "end of lifetime" (EOL) of the LD. For the violet LDs for example, it is set to 1.5 times the initial operating current of the LD.

To change limits, please perform as follows:

CMD>	!pass TuiOptics	
CMD>	sh limit	read out the actual setting of all limits
CMD>	limit warn <val>	<val> could be the old setting + 2mA
CMD>	limit soft <val>	<val> could be the old setting + 2mA
CMD>	limit clip <val>	<val> could be the old setting + 2mA
CMD>	save data	to store the new settings to the EEPROM

10.8 TEC Cooling Precautions for Non-Condensing Operation Conditions

The iBEAM smart is equipped with a TEC cooler to control and stabilize the laser diode temperature. In standard lasers, the laser diode and its mount are stabilized to 25 °C (77°F). This means, however, that especially in warm and humid surroundings, there could be condensation at the diode and the diode mount inside the iBEAM smart – this is a condition that has to be prevented. Please refer to the dew point table below to see if the environmental conditions are appropriate for the operation of the iBEAM smart.

The iBEAM smart laser diode temperature may vary depending on special customer requests. You can check the actual set point of the temperature by

- typing the command `sh temp` in a terminal software connected to the iBEAM smart. The iBEAM smart will then return the current laser diode temperature in °C.
- using the TOPAS iBeam smart software. The Temperature is displayed in the respective field in the Main tab.

The dew point table below indicates safe operating conditions for a standard laser operating at 25 °C diode temperature. Safe operating point means that the dew point is below the set laser diode temperature. For safety reasons, there should be a small margin of 2 °C, so the actual dew point must be below 23 °C. Grayed out cells show operating conditions where the laser may not be used, meaning all conditions, where the dew point is at 23 °C or higher.

Ambient Temperature in °C	15	20	25	30	35	40	45
Ambient Temperature in °F	59	68	77	86	95	104	113
Relative Ambient Humidity	Dew Points at corresponding Ambient Conditions						
90%	13,4	18,3	23,2	28,2	33,1	38,0	43,0
85%	12,5	17,4	22,3	27,2	32,1	37,0	41,9
80%	11,6	16,4	21,3	26,2	31,0	35,9	40,7
75%	10,6	15,4	20,3	25,1	29,9	34,7	39,5
70%	9,6	14,4	19,1	23,9	28,7	33,5	38,2
65%	8,5	13,2	18,0	22,7	27,4	32,2	36,9
60%	7,3	12,0	16,7	21,4	26,1	30,7	35,4
55%	6,0	10,7	15,3	20,0	24,6	29,2	33,9
50%	4,7	9,3	13,9	18,4	23,0	27,6	32,2
45%	3,2	7,7	12,2	16,8	21,3	25,8	30,3
40%	1,5	6,0	10,5	14,9	19,4	23,8	28,3
35%	-0,3	4,1	8,5	12,9	17,3	21,6	26,0
30%	-2,4	1,9	6,2	10,5	14,8	19,1	23,4
25%	-4,9	-0,6	3,6	7,8	12,0	16,2	20,4
20%	-7,8	-3,7	0,5	4,6	8,7	12,8	16,8

Table 1 Dew Point Table

10.9 Main Dimensions of iBEAM smart Laser Heads

10.9.1 iBEAM smart Laser Head

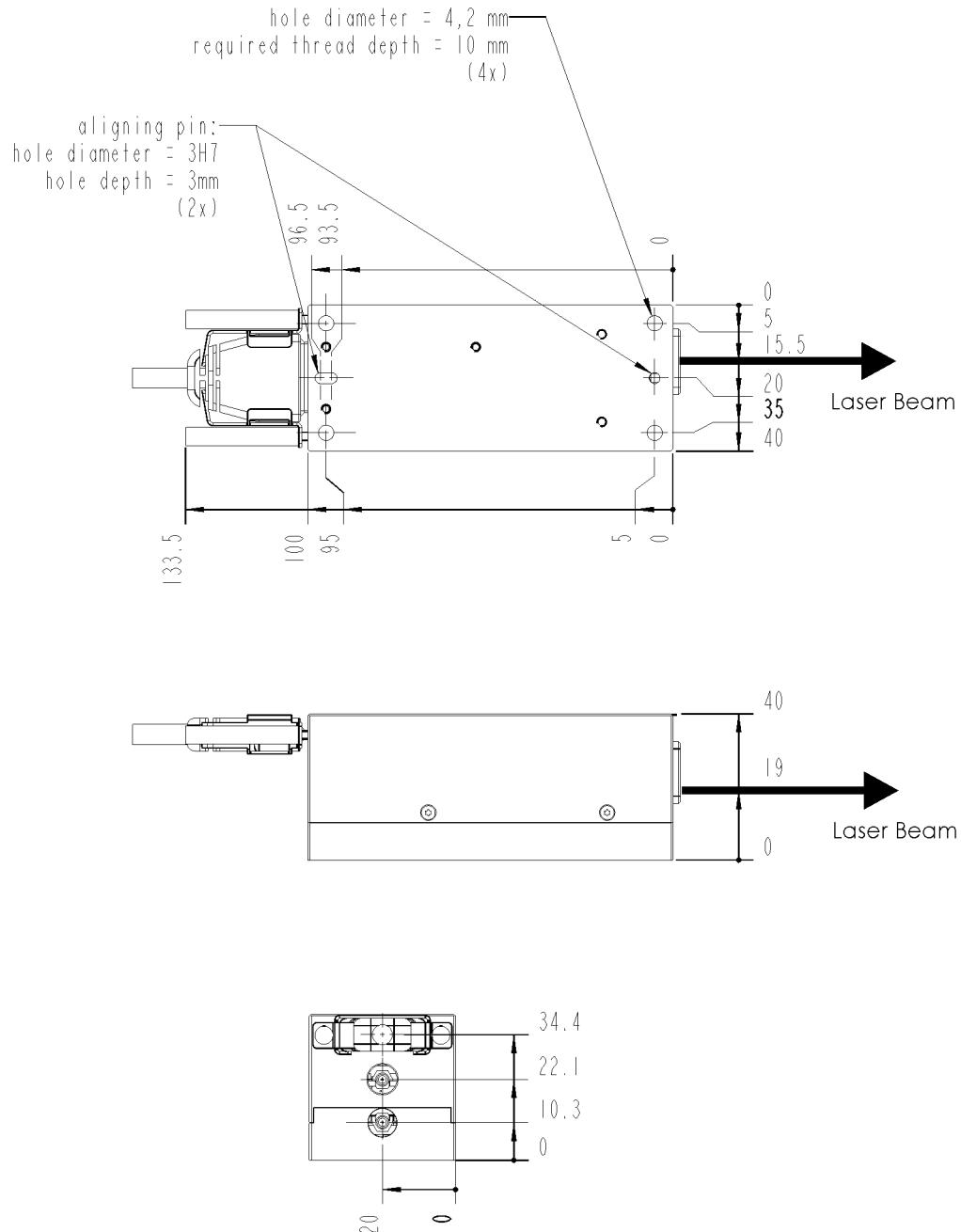


Figure 28 Main dimensions of iBEAM smart laser head

10.9.2 iBEAM smart Laser Head with SMARTDOCK

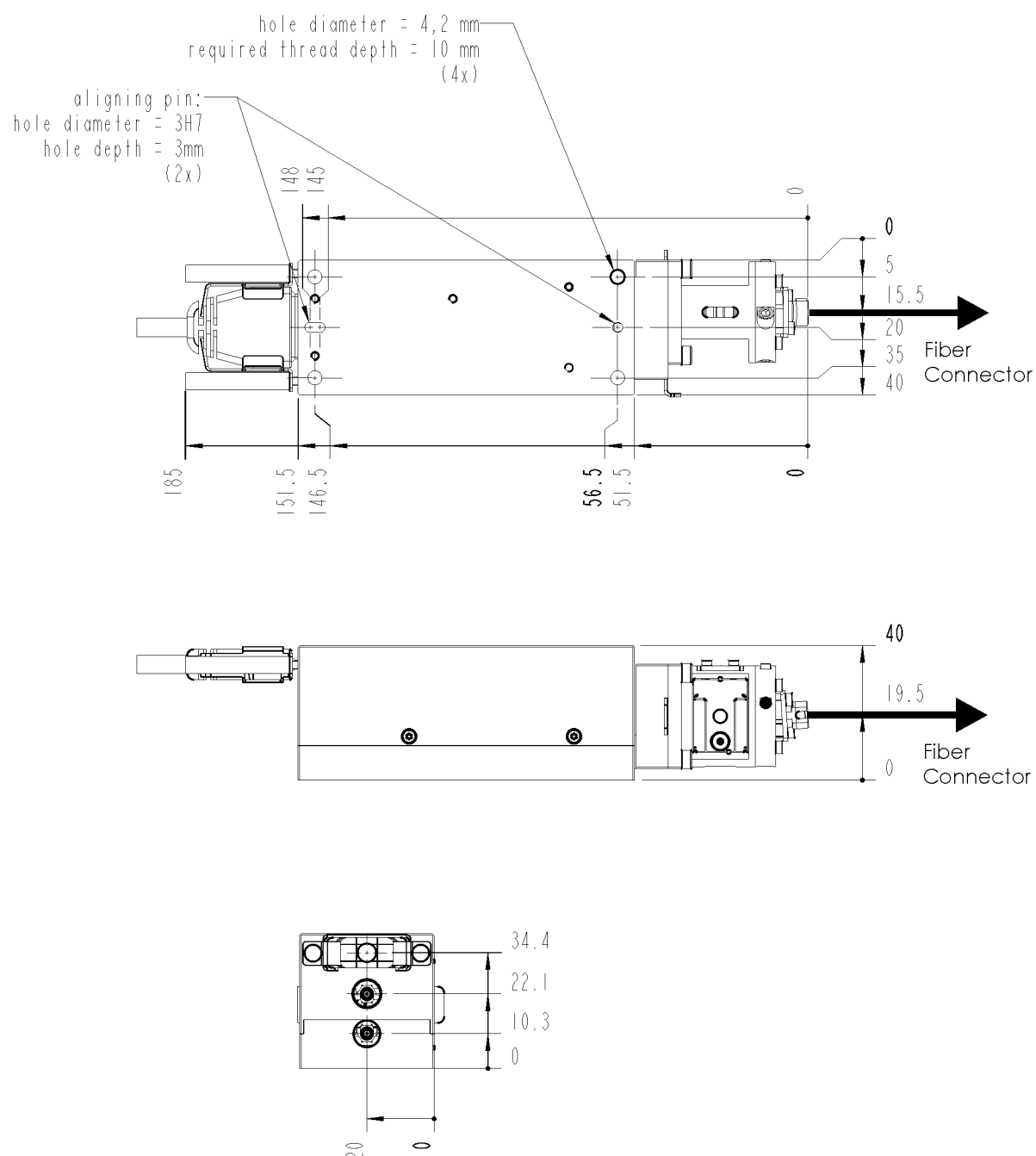


Figure 29 Main dimensions of iBEAM smart laser head with SMARTDOCK

10.9.3 Main Dimensions of iBEAM smart WS Laser Head

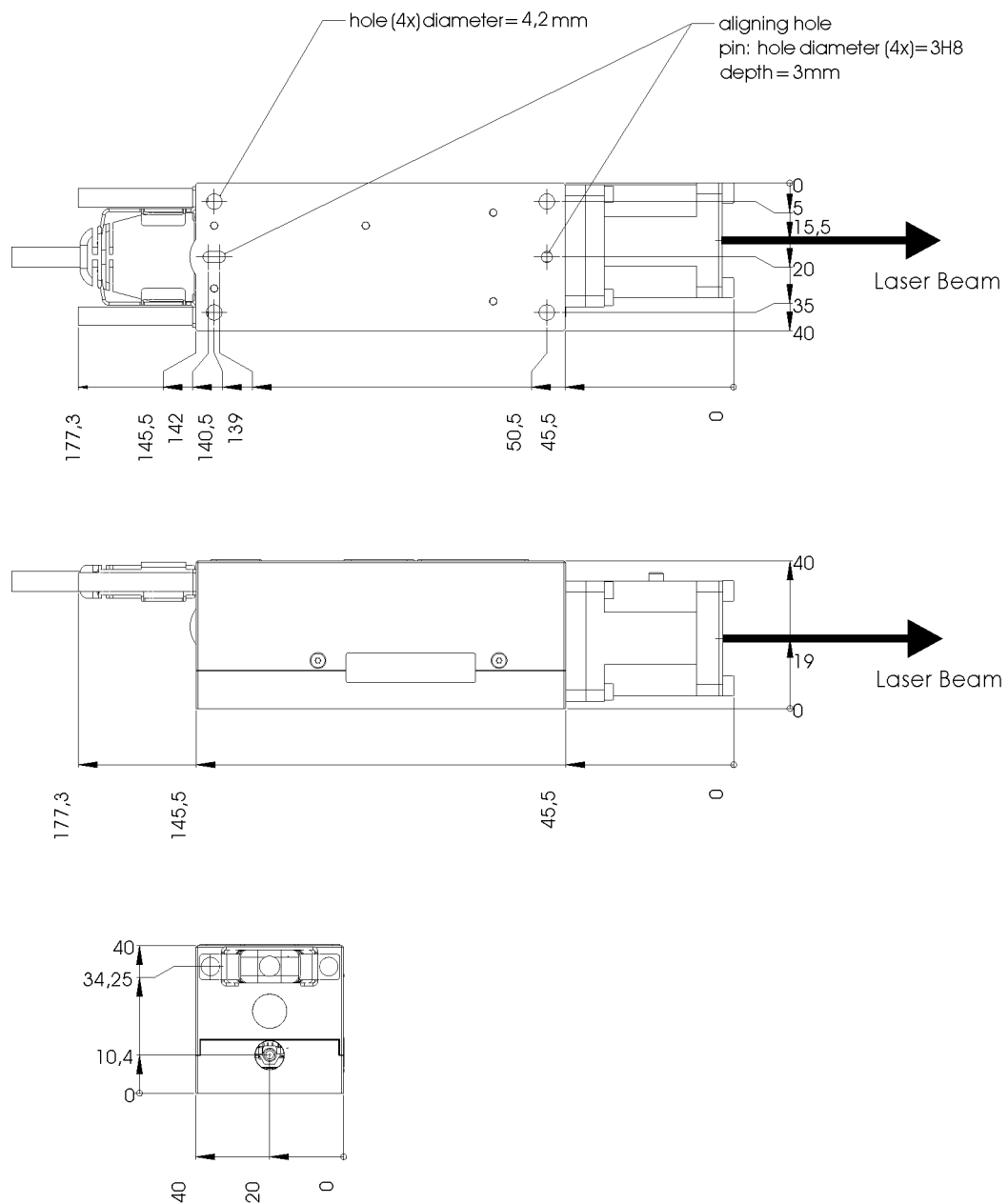


Figure 30 Main dimensions of iBEAM smart WS laser head

10.9.4 Main Dimensions of iBEAM smart WS Laser Head with SMARTDOCK

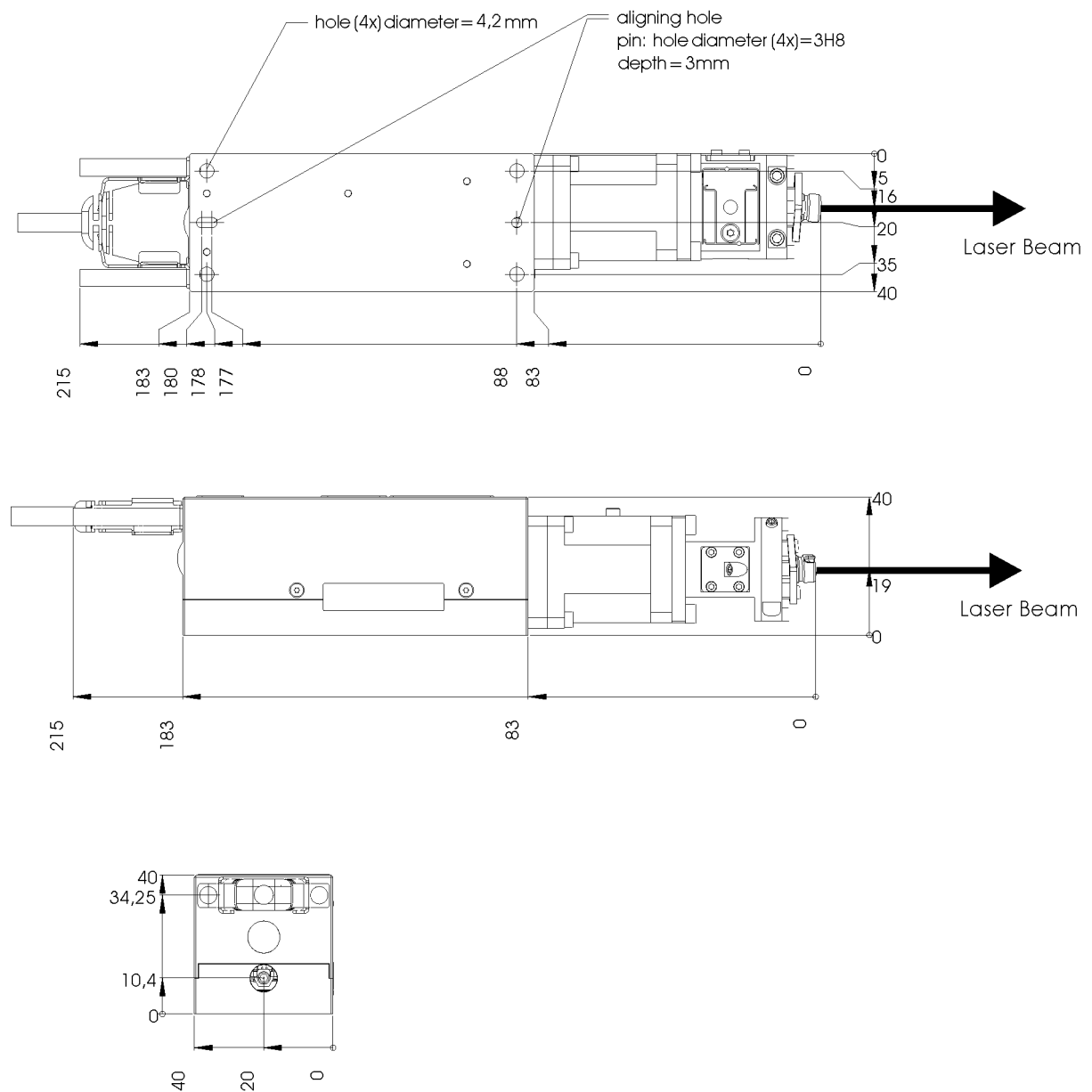


Figure 31 Main dimensions of iBEAM smart WS laser head with SMARTDOCK

10.10 Declaration of CE Conformity

Konformitätserklärung				
Declaration of Conformity / Declaration de Conformité				
QM-Formular:	F-173	Stand Formular:	17.02.2022	Version Formular:
			15	Seite:
				1 von 1

Wir / We / Nous
Anschrift / Address / Adresse

TOPTICA Photonics AG
Lochhamer Schlag 19
82166 Graefelfing
Germany



erklären in alleiniger Verantwortung, daß das Produkt / declare under our sole responsibility, that the product / déclarons sous notre seule responsabilité, que le produit

Bezeichnung / Name / Nom

Compact diode laser

Artikelnr. / Article No. / No. d'Article

iBeam smart, iB-SMART, iBeam smart PT

Beschreibung / Description / Description

Laserquelle auf Laserdioden-Basis / laser source based on laser diodes / laser basé sur diode laser

mit den grundlegenden Anforderungen der Richtlinien / fulfills the requirements of the standard and regulations of the directives / satisfait aux exigences des normes et directives

**2014/30/EU (Elektromagnetische Verträglichkeit), 2014/35/EU (Niederspannungsrichtlinie)
2011/65/EU and the amendment of Directive 2015/863 (RoHS-3 Richtlinie)**

übereinstimmt und damit den Bestimmungen entspricht. / and therefore corresponds to the regulations of the directive. / et, ainsi, correspond au règlements de la directive.

Angewendete harmonisierte Normen / Applied harmonized standards: / Normes harmonisées appliquées:

DIN EN 61326-1	VDE 0843-20-1:2013-07 Elektrische Mess-, Steuer-, Regel- und Laborgeräte - EMV-Anforderungen - Teil 1: Allgemeine Anforderungen (IEC 61326-1:2012); Deutsche Fassung EN 61326-1:2013 Electrical equipment for measurement, control and laboratory use - EMC requirements - Part 1: General requirements (IEC 61326-1:2012); German version EN 61326-1:2013
DIN EN 61010-1	Sicherheitsbestimmungen für elektrische Mess-, Steuer-, Regel- und Laborgeräte - Teil 1: Allgemeine Anforderungen (IEC 61010-1:2010 + COR:2011 + A1:2016, modifiziert + A1:2016/COR1:2019); Deutsche Fassung EN 61010-1:2010 + A1:2019 + A1:2019/AC:2019 Safety requirements for electrical equipment for measurement, control, and laboratory use - Part 1: General requirements (IEC 61010-1:2010 + COR:2011 + A1:2016, modified + A1:2016/COR1:2019); German version EN 61010-1:2010 + A1:2019 + A1:2019/AC:2019
DIN EN 60825-1	Sicherheit von Lasereinrichtungen - Teil 1: Klassifizierung von Anlagen und Anforderungen (IEC 60825-1:2014); Deutsche Fassung EN 60825-1:2014 + A11:2021 Safety of laser products - Part 1: Equipment classification and requirements (IEC 60825-1:2014); German version EN 60825-1:2014/A11_2021-02
DIN EN ISO 11252	Laser und Laseranlagen - Lasergerät - Mindestanforderungen an die Dokumentation (ISO 11252:2013); Deutsche Fassung EN ISO 11252:2013 Lasers and laser-related equipment- Laser device- Minimum requirements for documentation (ISO 11252:2013); German version EN ISO 11252:2013
DIN EN IEC 63000	Technische Dokumentation zur Beurteilung von Elektro- und Elektronikgeräten hinsichtlich der Beschränkung gefährlicher Stoffe (IEC 63000:2016); Deutsche Fassung EN IEC 63000:2018 Technical documentation for the assessment of electrical and electronic products with respect to the restriction of hazardous substances (IEC 63000:2016); German version EN IEC 63000:2018

Durch nicht von uns autorisierte Änderungen am Produkt verliert diese Erklärung ihre Gültigkeit. / Non authorized changes at the product result in the invalidity of this declaration. / Changements au produit, qui ne sont pas autorisées par nous, ont pour conséquence l'invalidité de cette déclaration.

Gräfelfing, den 26.02.2022

Dr. Wilhelm Kaenders
Vorstand

Ort und Datum der Ausstellung
Place and Date of Issue
Lieu et date d'établissement

Name und Unterschrift des Befugten
Name and Signature of authorized person
Nom et signature de la personne autorisée

10.11 EU Legislation for Electrical and Electronic Equipment (EEE)

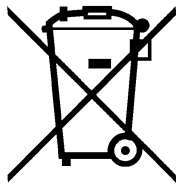
Companies selling electrical and electronic goods in the European Union must conform to the EU legislation for electrical and electronic equipment (EEE), which includes the Waste Electrical and Electronic Equipment Directive (WEEE). Assigned duties affect product design of the equipment, disposal of used appliances as well as organizational responsibilities, i.e. product registration.

There are different requirements for household WEEE and that which is sold business to business (B2B). All equipment TOPTICA Photonics SE handles is classed as B2B. TOPTICA is registered at the Competent Authority (Stiftung Elektro-Altgeräte Register EAR) under No. DE70442884.

At end-of life return your product back to TOPTICA. TOPTICA will dispose used equipment in such a manner as to meet all relevant local, country and EU requirements and guideline.

To return products please mark them clearly with "used device". Contact TOPTICA prior to shipping and send them to the following address:

TOPTICA Photonics SE
Lochhamer Schlag 19
D-82166 Graefelfing



The crossed out wheeled dust bin symbol indicates that products must be collected and disposed of separately from household waste.

TOPTICA reminds all end users of WEEE that you are responsible for deleting personal data on the waste equipment to be disposed of.

11 Guarantee and Service

As a first step toward obtaining technical support, please contact your local distributor or visit the support pages on our web site: <https://www.toptica.com/support/>.

In case you wish to return a product for diagnosis and/or repair, please contact us prior to sending it so we can issue a **Return Material Authorization** (RMA) number for you.

You can always contact TOPTICA service on the Internet by

- Email: service@toptica.com
- Web form: [service.toptica.com](https://www.toptica.com/service)

The service department at our headquarters in Munich can be reached by phone

- Tel: +49 89 85837150

For up-to-date phone numbers and addresses of our regional offices please refer to our web site: www.toptica.com/contact-us/

